MODULE 7 AND 8

1.Try Test-Connection and nslookup commands for below websites
www.google.com
www.facebook.com
www.amazon.com
www.github.com

Test-Connection does dns resolution first and does an icmp ping.

```
Windows PowerShell
PS C:\Users\yasha> Test-Connection www.google.com

Source        Destination     IPV4Address      IPV6Address
------        -----------     -----------      -----------
LAPTOP-5LJ... www.google.com  142.251.153.119  2404:6800:4002:818::2004
LAPTOP-5LJ... www.google.com  142.251.153.119  2404:6800:4002:818::2004
LAPTOP-5LJ... www.google.com  142.251.153.119  2404:6800:4002:818::2004
LAPTOP-5LJ... www.google.com  142.251.153.119  2404:6800:4002:818::2004


PS C:\Users\yasha> Test-Connection www.facebook.com

Source        Destination     IPV4Address      IPV6Address
------        -----------     -----------      -----------
LAPTOP-5LJ... www.facebook... 57.144.56.1      2a03:2880:f335:1:face:b00...
LAPTOP-5LJ... www.facebook... 57.144.56.1      2a03:2880:f335:1:face:b00...
LAPTOP-5LJ... www.facebook... 57.144.56.1      2a03:2880:f335:1:face:b00...
LAPTOP-5LJ... www.facebook... 57.144.56.1      2a03:2880:f335:1:face:b00...


PS C:\Users\yasha> Test-Connection www.amazon.com

Source        Destination     IPV4Address      IPV6Address
------        -----------     -----------      -----------
LAPTOP-5LJ... www.amazon.com  18.67.150.142    2600:140f:400:180::3bd4
LAPTOP-5LJ... www.amazon.com  18.67.150.142    2600:140f:400:180::3bd4
LAPTOP-5LJ... www.amazon.com  18.67.150.142    2600:9000:24d8:da00:7:49a...
LAPTOP-5LJ... www.amazon.com  18.67.150.142    2600:9000:24d8:da00:7:49a...


PS C:\Users\yasha> Test-Connection www.github.com

Source        Destination     IPV4Address      IPV6Address
------        -----------     -----------      -----------
LAPTOP-5LJ... www.github.com  20.207.73.82     64:ff9b::14cf:4952
LAPTOP-5LJ... www.github.com  20.207.73.82     64:ff9b::14cf:4952
LAPTOP-5LJ... www.github.com  20.207.73.82     64:ff9b::14cf:4952
LAPTOP-5LJ... www.github.com  20.207.73.82     64:ff9b::14cf:4952
```

```
PS C:\Users\yasha> Test-Connection www.cisco.com

Source        Destination     IPV4Address      IPV6Address
------        -----------     -----------      -----------
LAPTOP-5LJ... www.cisco.com   23.200.49.43     2600:140f:400:18e::b33
LAPTOP-5LJ... www.cisco.com   23.200.49.43     2600:140f:400:186::b33
LAPTOP-5LJ... www.cisco.com   23.200.49.43     2600:140f:400:186::b33
LAPTOP-5LJ... www.cisco.com   23.200.49.43     2600:140f:400:186::b33
```

Nslookup

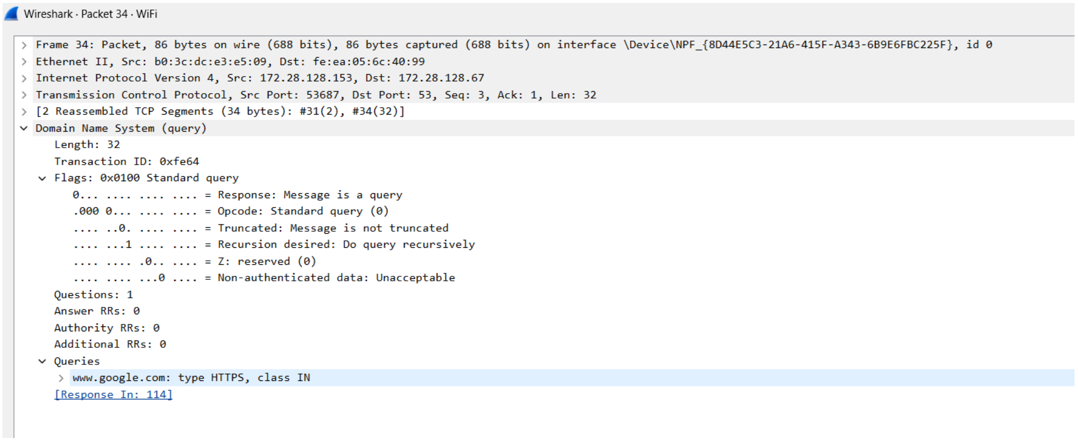
```
PS C:\Users\yasha> nslookup www.google.com
Server:  UnKnown
Address:  10.13.242.36

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:4826:7700::
          2001:4860:482c:7700::
          2001:4860:482a:7700::
          2001:4860:4827:7700::
          2001:4860:482d:7700::
          2001:4860:4828:7700::
          2001:4860:4829:7700::
          2001:4860:482b:7700::
          142.251.221.196

PS C:\Users\yasha> nslookup www.facebook.com
Server:  UnKnown
Address:  10.13.242.36

Non-authoritative answer:
Name:    star-mini.c10r.facebook.com
Addresses:  2a03:2880:f334:1:face:b00c:0:25de
          57.144.54.1
Aliases:  www.facebook.com
```

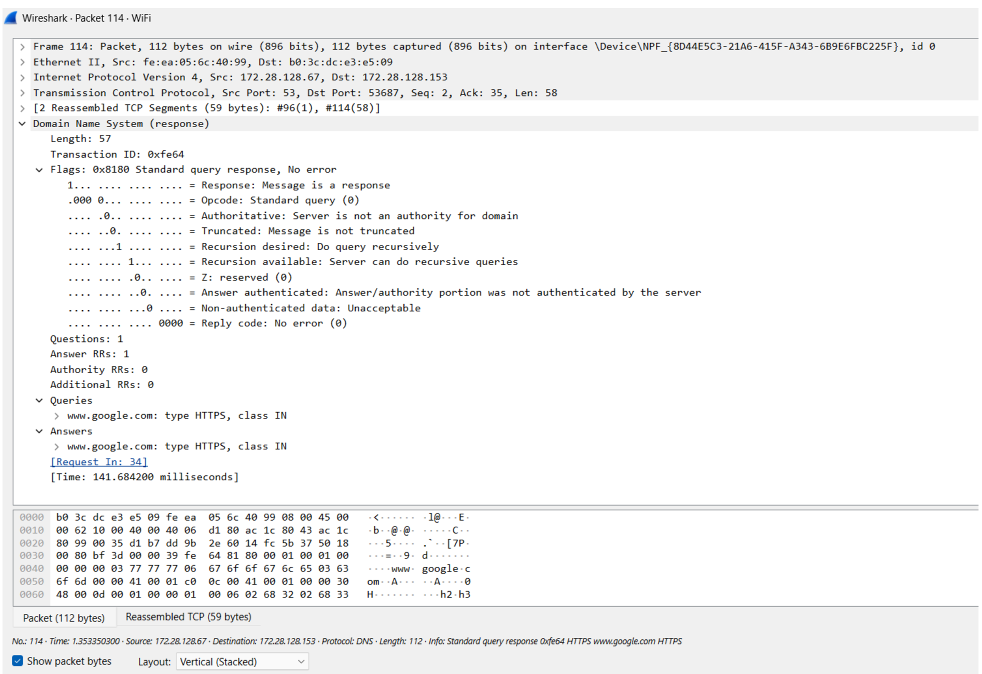
```
Windows PowerShell

PS C:\Users\yasha> nslookup www.facebook.com
Server:  UnKnown
Address:  10.13.242.36

Non-authoritative answer:
Name:    star-mini.c10r.facebook.com
Addresses:  2a03:2880:f334:1:face:b00c:0:25de
          57.144.54.1
Aliases:  www.facebook.com

PS C:\Users\yasha> nslookup www.amazon.com
Server:  UnKnown
Address:  10.13.242.36

Non-authoritative answer:
Name:    cf.47cf2c8c9-frontier.amazon.com
Addresses:  2600:9000:24d9:b000:7:49a5:5fd6:da1
          2600:9000:24d9:9c00:7:49a5:5fd6:da1
          2600:9000:24d9:7a00:7:49a5:5fd6:da1
          2600:9000:24d9:bc00:7:49a5:5fd6:da1
          2600:9000:24d9:de00:7:49a5:5fd6:da1
          2600:9000:24d9:3000:7:49a5:5fd6:da1
          2600:9000:24d9:1c00:7:49a5:5fd6:da1
          2600:9000:24d9:4a00:7:49a5:5fd6:da1
          18.67.150.142
Aliases:  www.amazon.com
          tp.47cf2c8c9-frontier.amazon.com
```

```
PS C:\Users\yasha> nslookup www.github.com
Server:  UnKnown
Address:  10.13.242.36

Non-authoritative answer:
Name:    github.com
Addresses:  64:ff9b::14cf:4952
          20.207.73.82
Aliases:  www.github.com

PS C:\Users\yasha> nslookup www.cisco.com
Server:  UnKnown
Address:  10.13.242.36

Non-authoritative answer:
Name:    e2867.dsca.akamaiedge.net
Addresses:  2600:140f:400:186::b33
          2600:140f:400:181::b33
          23.200.49.43
Aliases:  www.cisco.com
          www.cisco.com.akadns.net
          wwwds.cisco.com.edgekey.net
          wwwds.cisco.com.edgekey.net.globalredir.akadns.net

PS C:\Users\yasha>
```

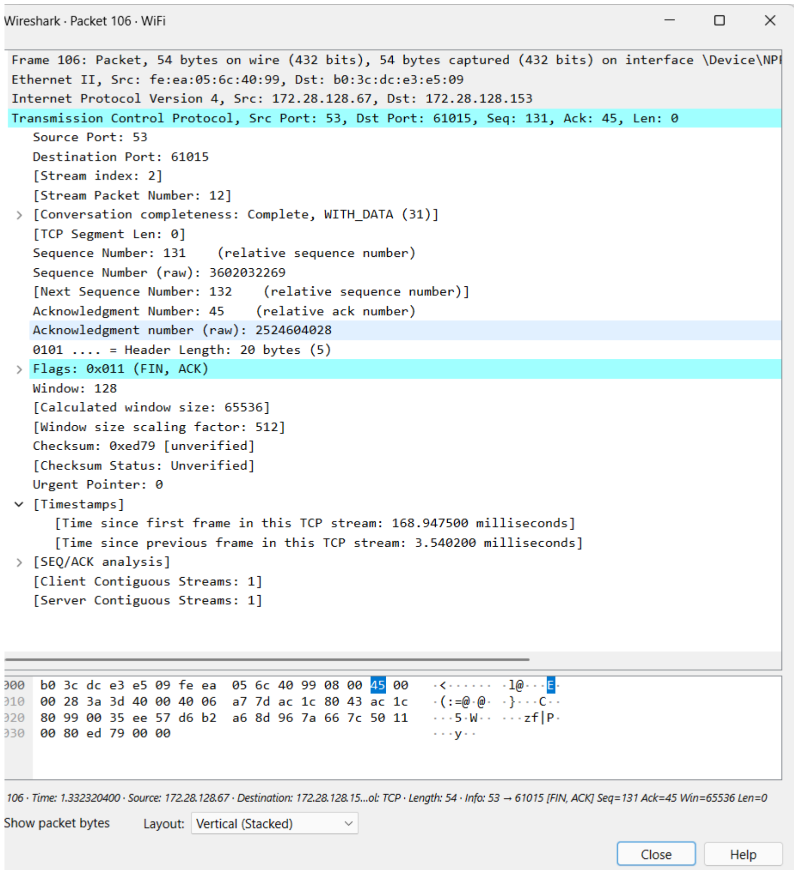
2.        Use Wireshark to capture and analyze DNS, TCP, UDP traffic and packet header, packet flow, options and flags.

DNS

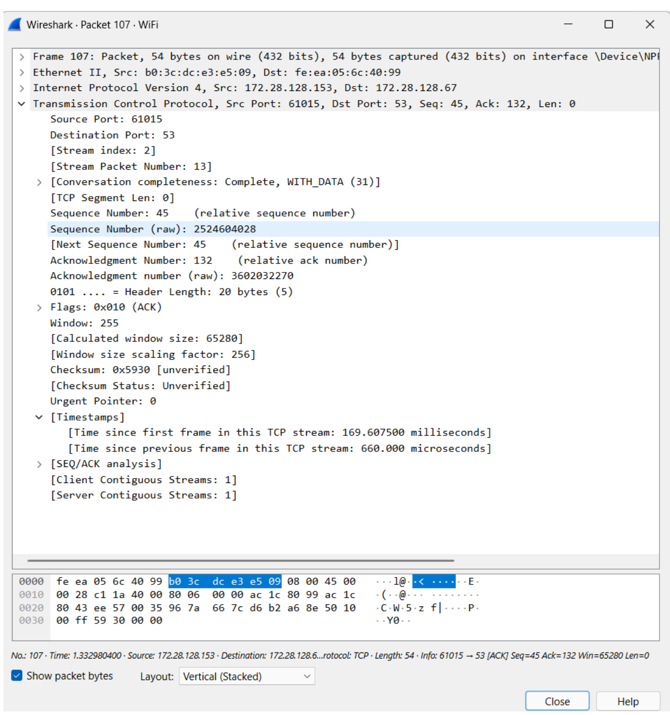


# TCP

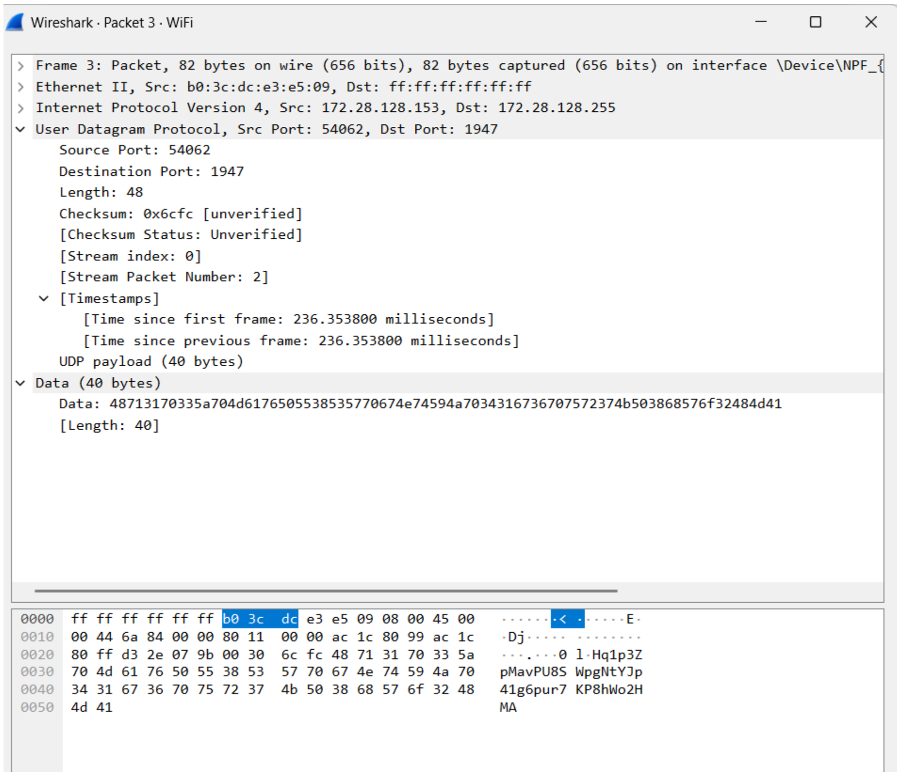


Wireshark · Packet 106 · WiFi
Frame 106: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPI
Ethernet II, Src: fe:ea:05:6c:40:99, Dst: b0:3c:dc:e3:e5:09
Internet Protocol Version 4, Src: 172.28.128.67, Dst: 172.28.128.153
Transmission Control Protocol, Src Port: 53, Dst Port: 61015, Seq: 131, Ack: 45, Len: 0
Source Port: 53
Destination Port: 61015
[Stream index: 2]
[Stream Packet Number: 12]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 131 (relative sequence number)
Sequence Number (raw): 3602032269
[Next Sequence Number: 132 (relative sequence number)]
Acknowledgment Number: 45 (relative ack number)
Acknowledgment number (raw): 2524604028
0101 .... = Header Length: 20 bytes (5)
Flags: 0x011 (FIN, ACK)
Window: 128
[Calculated window size: 65536]
[Window size scaling factor: 512]
Checksum: 0xed79 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[Time since first frame in this TCP stream: 168.947500 milliseconds]
[Time since previous frame in this TCP stream: 3.540200 milliseconds]
[SEQ/ACK analysis]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
000 b0 3c dc e3 e5 09 fe ea 05 6c 40 99 08 00 45 00
010 00 28 3a 3d 40 00 40 06 a7 7d ac 1c 80 43 ac 1c
020 80 99 00 35 ee 57 d6 b2 a6 8d 96 7a 66 7c 50 11
030 00 80 ed 79 00 00
106 · Time: 1.332320400 · Source: 172.28.128.67 · Destination: 172.28.128.15...ol: TCP · Length: 54 · Info: 53 → 61015 [FIN, ACK] Seq=131 Ack=45 Win=65536 Len=0
Show packet bytes
Layout: Vertical (Stacked)
Close
Help

Wireshark · Packet 107 · WiFi
Frame 107: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPI
Ethernet II, Src: b0:3c:dc:e3:e5:09, Dst: fe:ea:05:6c:40:99
Internet Protocol Version 4, Src: 172.28.128.153, Dst: 172.28.128.67
Transmission Control Protocol, Src Port: 61015, Dst Port: 53, Seq: 45, Ack: 132, Len: 0
Source Port: 61015
Destination Port: 53
[Stream index: 2]
[Stream Packet Number: 13]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 45 (relative sequence number)
Sequence Number (raw): 2524604028
[Next Sequence Number: 45 (relative sequence number)]
Acknowledgment Number: 132 (relative ack number)
Acknowledgment number (raw): 3602032270
0101 .... = Header Length: 20 bytes (5)
Flags: 0x010 (ACK)
Window: 255
[Calculated window size: 65280]
[Window size scaling factor: 256]
Checksum: 0x5930 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[Time since first frame in this TCP stream: 169.607500 milliseconds]
[Time since previous frame in this TCP stream: 660.000 microseconds]
[SEQ/ACK analysis]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
0000 fe ea 05 6c 40 99 b0 3c dc e3 e5 09 08 00 45 00
0010 00 28 c1 1a 40 00 80 06 00 00 ac 1c 80 99 ac 1c
0020 80 43 ee 57 00 35 96 7a 66 7c d6 b2 a6 8e 50 10
0030 00 ff 59 30 00 00
No.: 107 · Time: 1.332980400 · Source: 172.28.128.153 · Destination: 172.28.128.6...rotocol: TCP · Length: 54 · Info: 61015 → 53 [ACK] Seq=45 Ack=132 Win=65280 Len=0
Show packet bytes
Layout: Vertical (Stacked)
Close
Help

UDP




3. Explore traceroute/tracert for different websites eg:google.com and analyse the parameters in the output and explore different options for traceroute command

The tracert command helps identify the path taken by packets across a network and diagnose latency or connectivity issues. By analyzing hop count, delay, and timeouts, network performance and routing problems can be understood.

- Initial hops belong to local network (router, ISP)
- Middle hops represent intermediate ISP routers
- Final hop is destination server
- Delay increases as distance increases
- Uses ICMP Echo Requests
- Sends packets with increasing TTL (Time To Live)
- Each router decreases TTL by 1
- When TTL = 0, router sends back a response
- This helps identify each hop in the path

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8039]
(c) Microsoft Corporation. All rights reserved.

C:\Users\yasha>3. Explore traceroute/tracert for different websites eg:google.com and ar
ut and explore different options for traceroute command
'3.' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\yasha>tracert google.com

Tracing route to google.com [2404:6800:4007:832::200e]
over a maximum of 30 hops:

  1     3 ms     2 ms     1 ms  2409:40f4:a5:aa2d::fe
  2     *        *        *     Request timed out.
  3   247 ms    13 ms    17 ms  2405:200:5218:21:3925::1
  4   238 ms    13 ms    17 ms  2405:200:88c:1513:62::6
  5     *        *        *     Request timed out.
  6   243 ms    19 ms    17 ms  2405:200:801:900::1626
  7     *        *        *     Request timed out.
  8   244 ms    16 ms    78 ms  2001:4860:1:1::170
  9   227 ms    19 ms    20 ms  2001:4860:1:1::170
 10    40 ms    72 ms   243 ms  2404:6800:8202:280::1
 11   135 ms   232 ms   244 ms  2001:4860:0:1::5658
 12   249 ms   224 ms   238 ms  2001:4860:0:1::34c9
 13   252 ms    17 ms    18 ms  pnmaaa-az-in-x0e.1e100.net [2404:6800:4007:832::200e]

Trace complete.
```

Parameters in Output

- Hop Number (1, 2, 3, …)

Each hop represents a router between source and destination.

- Round Trip Time (ms)

Three time values are shown for each hop, indicating delay for three probe packets.

- IP Address / Hostname

Shows the router or device at that hop.

- * (Asterisk)

Indicates timeout or no response from that hop.

Common tracert Options

-d (Do not resolve hostnames)
tracert -d google.com

- Faster output (skips DNS lookup)

```
C:\Users\yasha>tracert -d google.com

Tracing route to google.com [2404:6800:4007:832::200e]
over a maximum of 30 hops:

  1     2 ms     3 ms     2 ms  2409:40f4:a5:aa2d::fe
  2     *        *        *     Request timed out.
  3    61 ms    27 ms    19 ms  2405:200:5218:21:3925::1
  4    41 ms    18 ms    17 ms  2405:200:88c:1513:62::6
  5     *        *        *     Request timed out.
  6   150 ms    13 ms    13 ms  2405:200:801:900::1626
  7     *        *        *     Request timed out.
  8   261 ms    35 ms   287 ms  2001:4860:1:1::170
  9   250 ms    21 ms    28 ms  2001:4860:1:1::170
 10    46 ms    74 ms    17 ms  2404:6800:8202:280::1
 11   239 ms    13 ms    18 ms  2001:4860:0:1::5658
 12    46 ms    23 ms    15 ms  2001:4860:0:1::34c9
 13   246 ms    11 ms    20 ms  2404:6800:4007:832::200e
```

-h (Maximum hops)
tracert -h 10 google.com

- Limits number of hops to 10

```
C:\Users\yasha>tracert -h 10 google.com

Tracing route to google.com [2404:6800:4007:836::200e]
over a maximum of 10 hops:

  1     1 ms     1 ms     1 ms  2409:40f4:a5:aa2d::fe
  2     *        *        *     Request timed out.
  3    42 ms    27 ms    13 ms  2405:200:5218:21:3925::1
  4    43 ms    18 ms    18 ms  2405:200:88c:1513:62::6
  5     *        *        *     Request timed out.
  6   236 ms    14 ms    15 ms  2405:200:801:900::1628
  7     *        *        *     Request timed out.
  8    79 ms    35 ms    32 ms  2001:4860:1:1::170
  9    45 ms    15 ms   236 ms  2001:4860:1:1::170
 10    21 ms    15 ms    24 ms  2404:6800:8201:380::1

Trace complete.
```

-w (Timeout)
tracert -w 1000 google.com

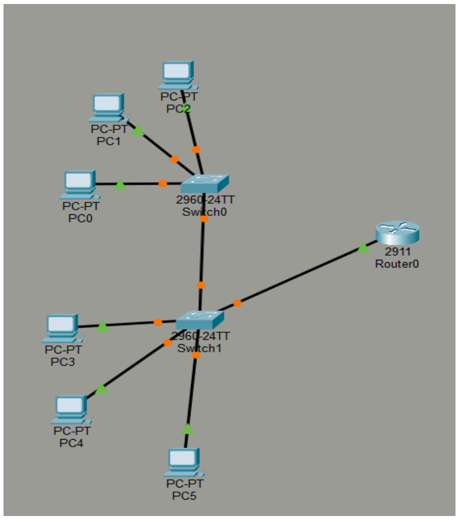
- Wait time in milliseconds for each reply

```
C:\Users\yasha>tracert -w 1000 google.com

Tracing route to google.com [2404:6800:4007:836::200e]
over a maximum of 30 hops:

  1   222 ms     2 ms     3 ms  2409:40f4:a5:aa2d::fe
  2     *        *        *     Request timed out.
  3   243 ms    28 ms    17 ms  2405:200:5218:21:3925::1
  4   245 ms    11 ms    15 ms  2405:200:88c:1513:62::6
  5     *        *        *     Request timed out.
  6    62 ms    21 ms    23 ms  2405:200:801:900::1628
  7     *        *        *     Request timed out.
  8   228 ms   217 ms   243 ms  2001:4860:1:1::170
  9   200 ms   223 ms   228 ms  2001:4860:1:1::170
 10   245 ms    17 ms   238 ms  2404:6800:8201:380::1
 11   104 ms    22 ms    13 ms  2001:4860:0:1::559a
 12    22 ms    37 ms    28 ms  2001:4860:0:1::55d3
 13   235 ms   195 ms   224 ms  lcmaaa-az-in-x0e.1e100.net [2404:6800:4007:836::200e]

Trace complete.
```

-4 (IPv4)
tracert -4 [google.com](google.com)

```
C:\Users\yasha>tracert -4 google.com

Tracing route to google.com [142.251.221.110]
over a maximum of 30 hops:

  1     1 ms     1 ms     2 ms  172.28.128.67
  2     *        *        *     Request timed out.
  3     *        *        *     Request timed out.
  4     *        *        *     Request timed out.
  5   179 ms    18 ms   251 ms  172.27.251.98
  6   240 ms    19 ms    18 ms  192.168.142.20
  7     *        *        *     Request timed out.
  8     *        *        *     Request timed out.
  9    46 ms    23 ms    37 ms  209.85.175.48
 10    46 ms    18 ms    16 ms  216.239.51.91
 11   257 ms    27 ms   224 ms  142.251.55.69
 12   179 ms    22 ms    25 ms  cgk03s03-in-f14.1e100.net [142.251.221.110]

Trace complete.
```

4. Set up trunk ports between switches and try pinging between different vlans




vlan configuration on switch0

```
Switch#
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

```
%SYS-5-CONFIG_I: Configured from console by console

Switch#show interface status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    10         a-full  a-100 10/100BaseTX
Fa0/2                        connected    10         a-full  a-100 10/100BaseTX
Fa0/3                        connected    10         a-full  a-100 10/100BaseTX
```

```
Switch>enable
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
10   IT                               active    Fa0/1, Fa0/2, Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Setting up the trunk port

```
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface Fa0/24
Switch(config-if)#switchport mode trunk
```

```
Fa0/24                    connected    trunk      a-full  a-100 10/100BaseTX
```

Vlan configuration on switch1

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
20   Fin                              active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface Fa0/24
Switch(config-if)#switchport  mode trunk
Switch(config-if)#exit
Switch(config)#
Switch(config)#exit
```

```
Fa0/24                    connected    trunk      a-full  a-100 10/100BaseTX
```

Enabling the interface in router

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

Router on stick architecture configuring the interfaces for vlan

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#interface g0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to
up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#exit
Router(config)#interface g0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to
up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
```

On switch1 shows the trunked ports and vlans allowed on those trunked ports

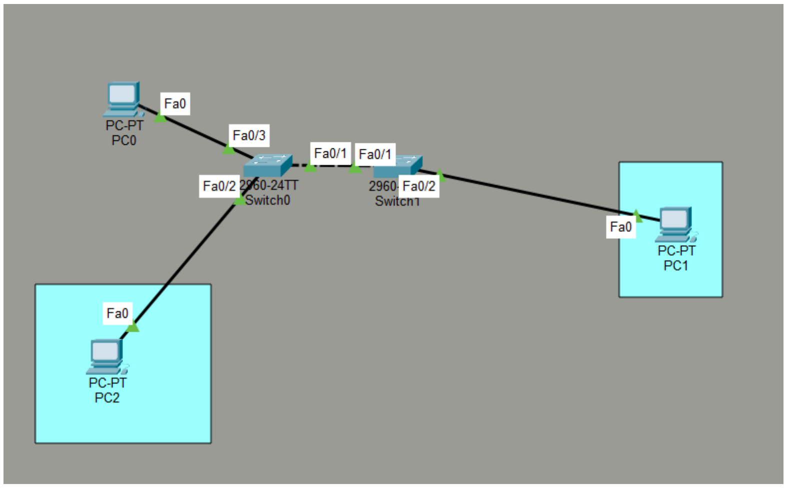
```
Switch1

Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/24      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005
Fa0/24      1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20
Fa0/24      1,10,20

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20
Fa0/24      1,10,20

Switch#
```

Pinging pcs in two different [vlans. It](#) has to be routed via the routers.

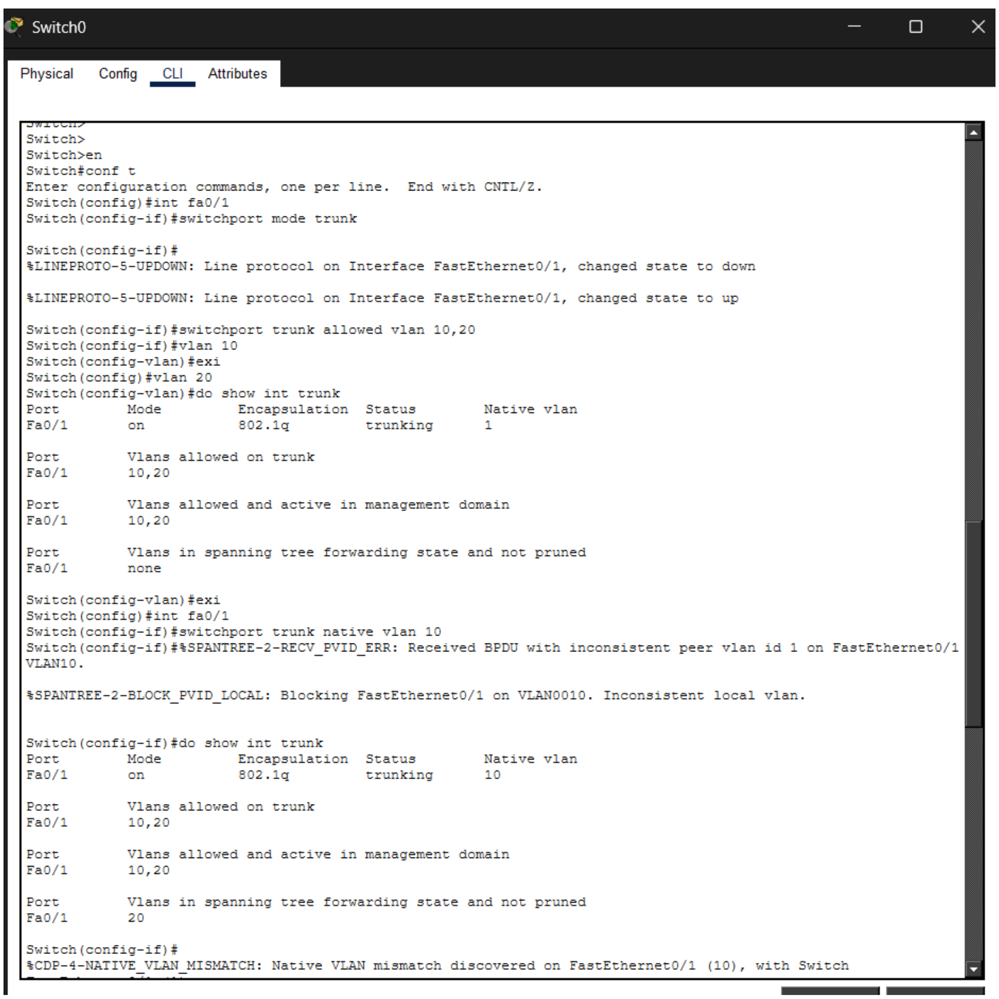
```
PC0

Physical  Config  Desktop  Programming  Attributes

Command Prompt                                                X

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

5. Change the native VLAN on a trunk port.Test for VLAN mismatches and Troubleshoot.

Network topology




Configuring the switches to have a native vlan mismatch



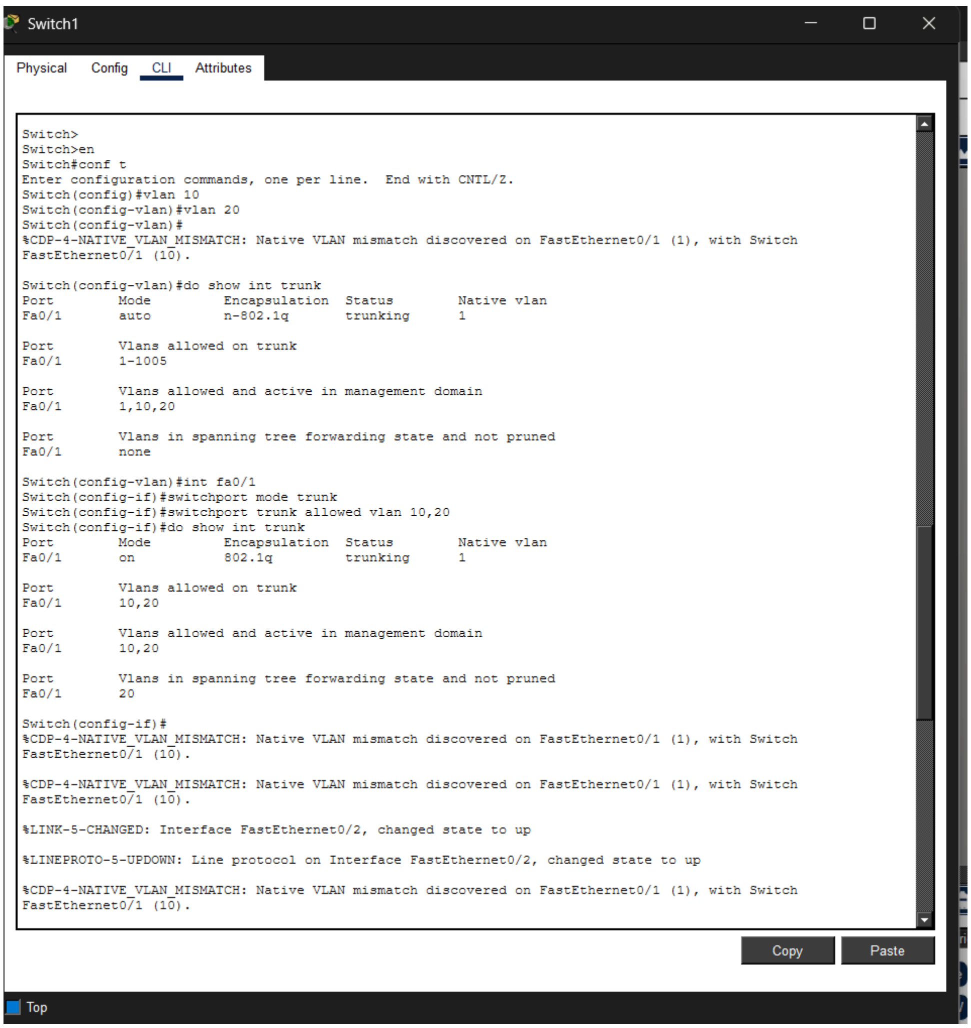

```
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#vlan 10
Switch(config-vlan)#exi
Switch(config)#vlan 20
Switch(config-vlan)#do show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       10,20

Port        Vlans allowed and active in management domain
Fa0/1       10,20

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       none

Switch(config-vlan)#exi
Switch(config)#int fa0/1
Switch(config-if)#switchport trunk native vlan 10
Switch(config-if)#%SPANTREE-2-RECV_PVID_ERR: Received BPDU with inconsistent peer vlan id 1 on FastEthernet0/1
VLAN10.

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/1 on VLAN0010. Inconsistent local vlan.


Switch(config-if)#do show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      10

Port        Vlans allowed on trunk
Fa0/1       10,20

Port        Vlans allowed and active in management domain
Fa0/1       10,20

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       20

Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (10), with Switch
```

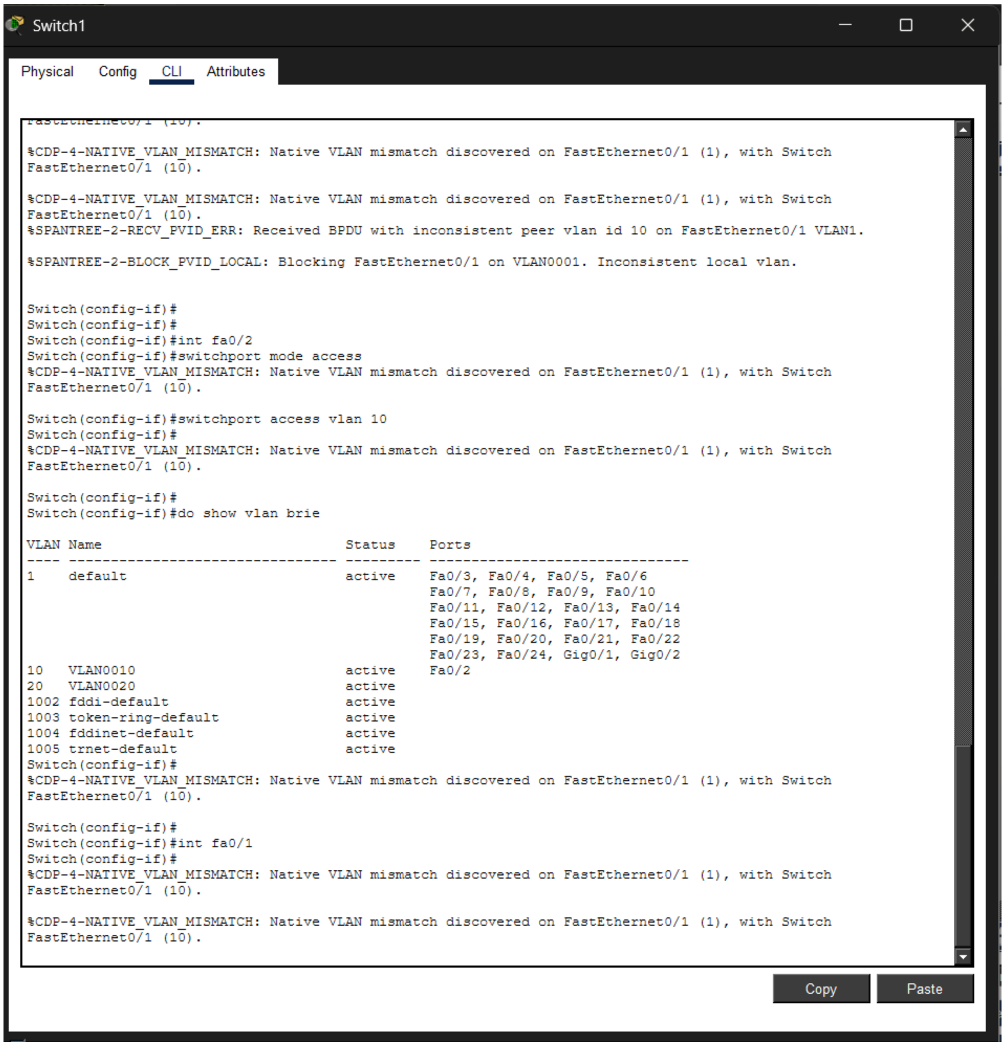
Switch0

Physical Config CLI Attributes

```
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (10), with Switch
FastEthernet0/1 (1).

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (10), with Switch
FastEthernet0/1 (1).

Switch(config-if)#
Switch(config-if)#
Switch(config-if)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#show vlan bri
                    ^
% Invalid input detected at '^' marker.

Switch(config-if)#do show vlan bri

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
10   VLAN0010                         active    Fa0/2
20   VLAN0020                         active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (10), with Switch
FastEthernet0/1 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (10), with Switch
FastEthernet0/1 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (10), with Switch
FastEthernet0/1 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (10), with Switch
FastEthernet0/1 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (10), with Switch
FastEthernet0/1 (1).
```

SW1


Switch1
Physical
Config
CLI
Attributes
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#vlan 20
Switch(config-vlan)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch FastEthernet0/1 (10).
Switch(config-vlan)#do show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       auto         n-802.1q       trunking      1
Port        Vlans allowed on trunk
Fa0/1       1-1005
Port        Vlans allowed and active in management domain
Fa0/1       1,10,20
Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       none
Switch(config-vlan)#int fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#do show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Port        Vlans allowed on trunk
Fa0/1       10,20
Port        Vlans allowed and active in management domain
Fa0/1       10,20
Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       20
Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch FastEthernet0/1 (10).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch FastEthernet0/1 (10).
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with Switch FastEthernet0/1 (10).
Copy
Paste
Top



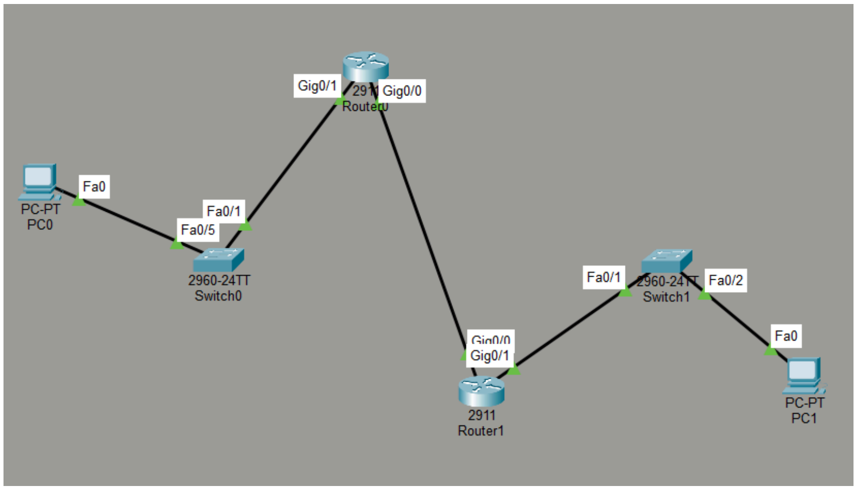

Native VLAN is VLAN for untagged traffic.

Mismatch Causes:

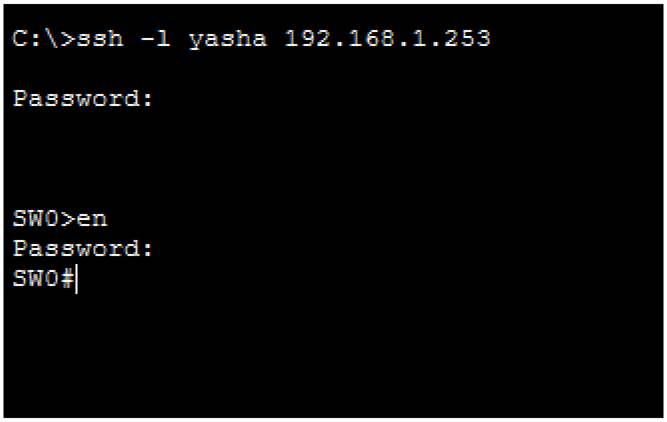
- Traffic leaks
- VLAN hopping risk
- Misdelivery

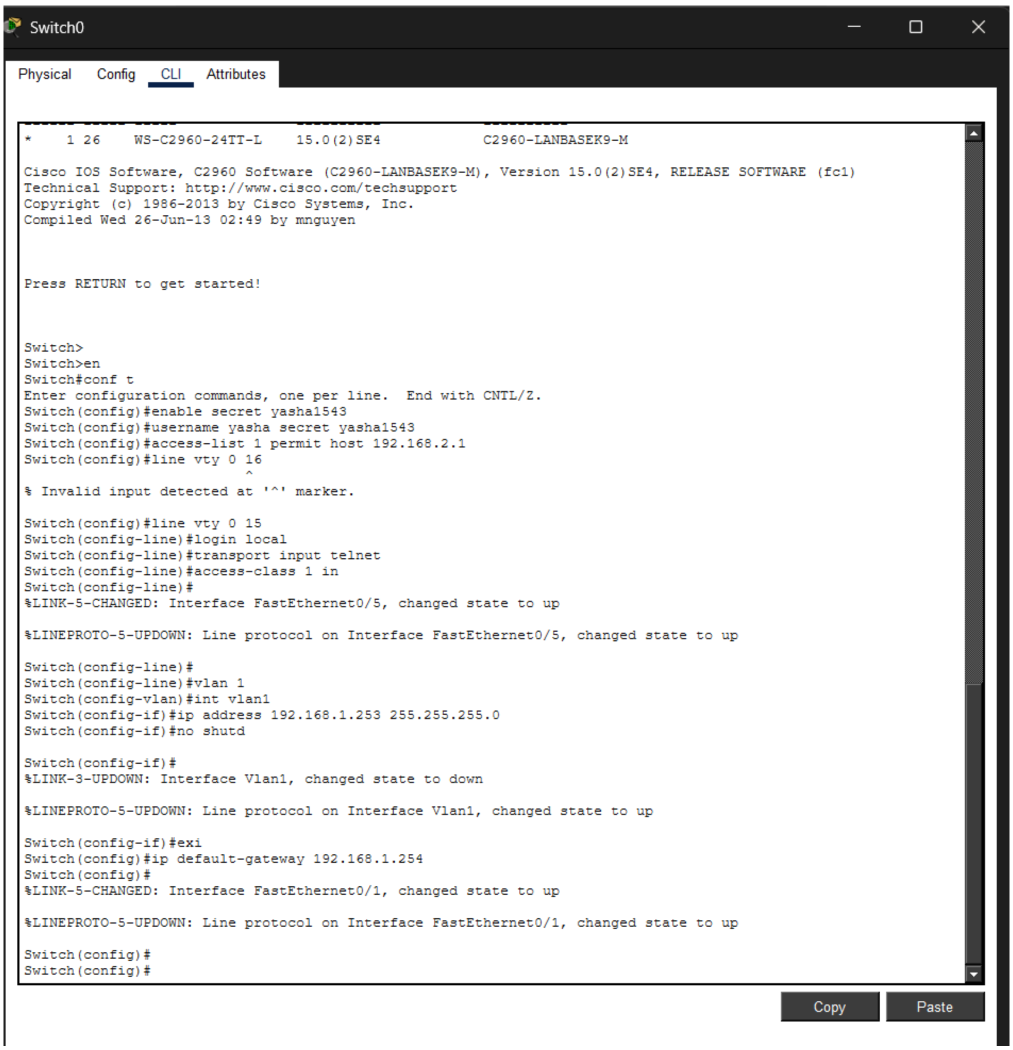
6. Configuring the SVI interface so that the pc can access the switch remotely using telnet or ssh.

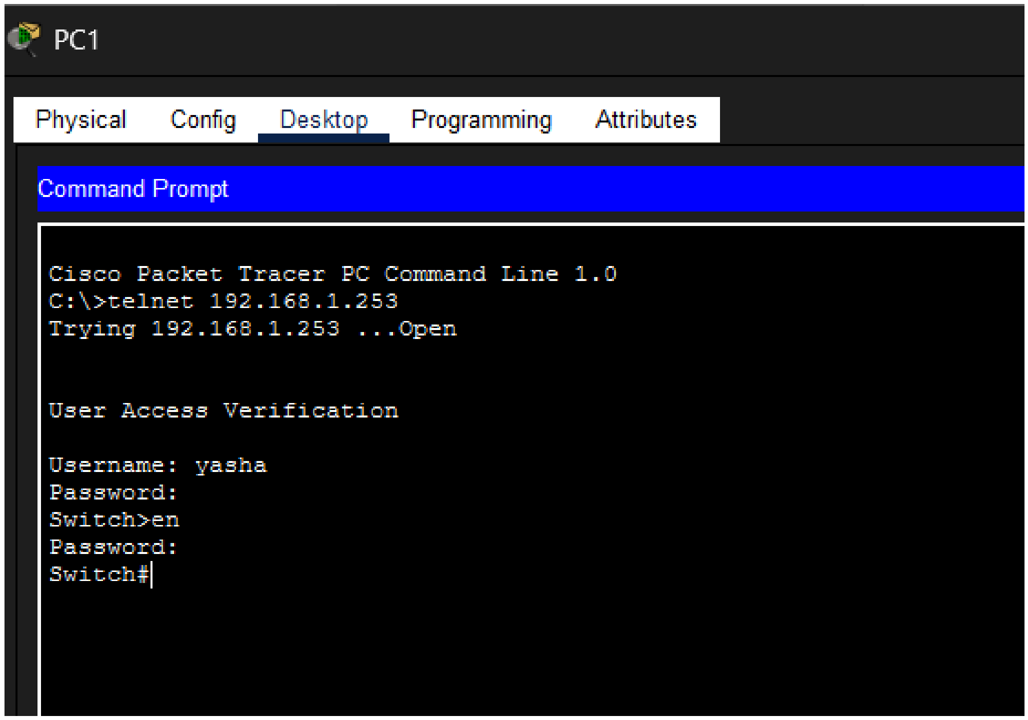
7. Configuring the SVI interface so that the pc can access the switch remotely using telnet or ssh.




```
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#hostname SW0
SW0(config)#ip domain name yasha.com
SW0(config)#crypto key generate rsa
The name for the keys will be: SW0.yasha.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

SW0(config)#
*Mar 1 0:25:9.41: %SSH-5-ENABLED: SSH 1.99 has been enabled
SW0(config)#do show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
SW0(config)#transport input ssh
              ^
% Invalid input detected at '^' marker.

SW0(config)#line vty 0 15
SW0(config-line)#login local
SW0(config-line)#transport input ssh
SW0(config-line)#access-class 1 in
SW0(config-line)#
```

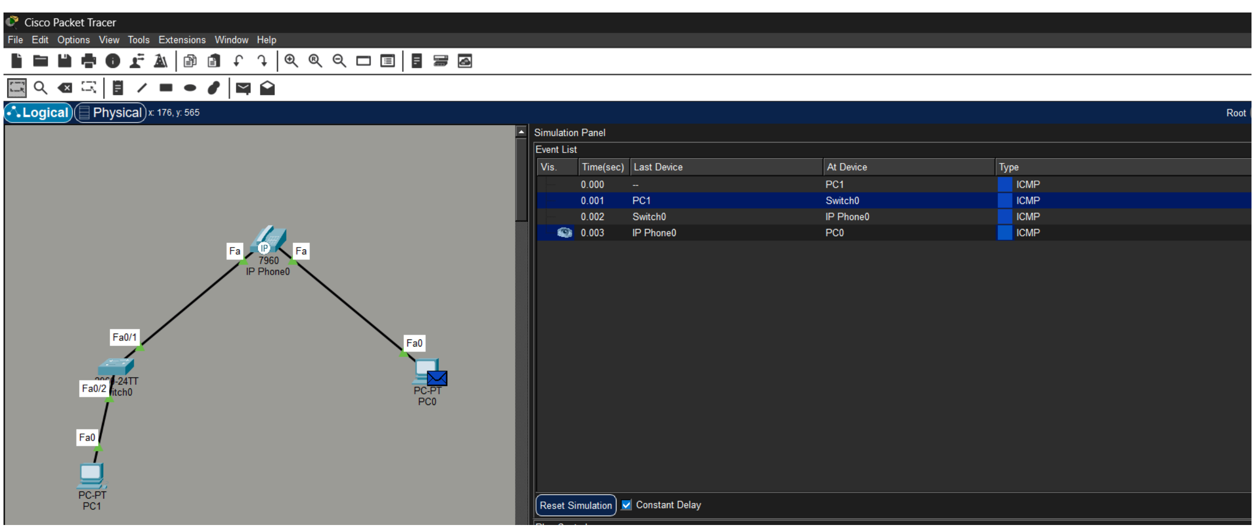

C:\>ssh -l yasha 192.168.1.253
Password:
SW0>en
Password:
SW0#


Switch0
Physical
Config
CLI
Attributes
*    1 26    WS-C2960-24TT-L    15.0(2)SE4           C2960-LANBASEK9-M
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen
Press RETURN to get started!
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#enable secret yasha1543
Switch(config)#username yasha secret yasha1543
Switch(config)#access-list 1 permit host 192.168.2.1
Switch(config)#line vty 0 16
^
% Invalid input detected at '^' marker.
Switch(config)#line vty 0 15
Switch(config-line)#login local
Switch(config-line)#transport input telnet
Switch(config-line)#access-class 1 in
Switch(config-line)#
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
Switch(config-line)#
Switch(config-line)#vlan 1
Switch(config-vlan)#int vlan1
Switch(config-if)#ip address 192.168.1.253 255.255.255.0
Switch(config-if)#no shutd
Switch(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
Switch(config-if)#exi
Switch(config)#ip default-gateway 192.168.1.254
Switch(config)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Switch(config)#
Switch(config)#
Copy
Paste




7. You have a Cisco switch and a VoIP phone that needs to be placed in a voice VLAN (VLAN 20). The data for the PC should remain in a separate VLAN (VLAN 10). Configure the switch port to support both voice and data traffic.

Network topology

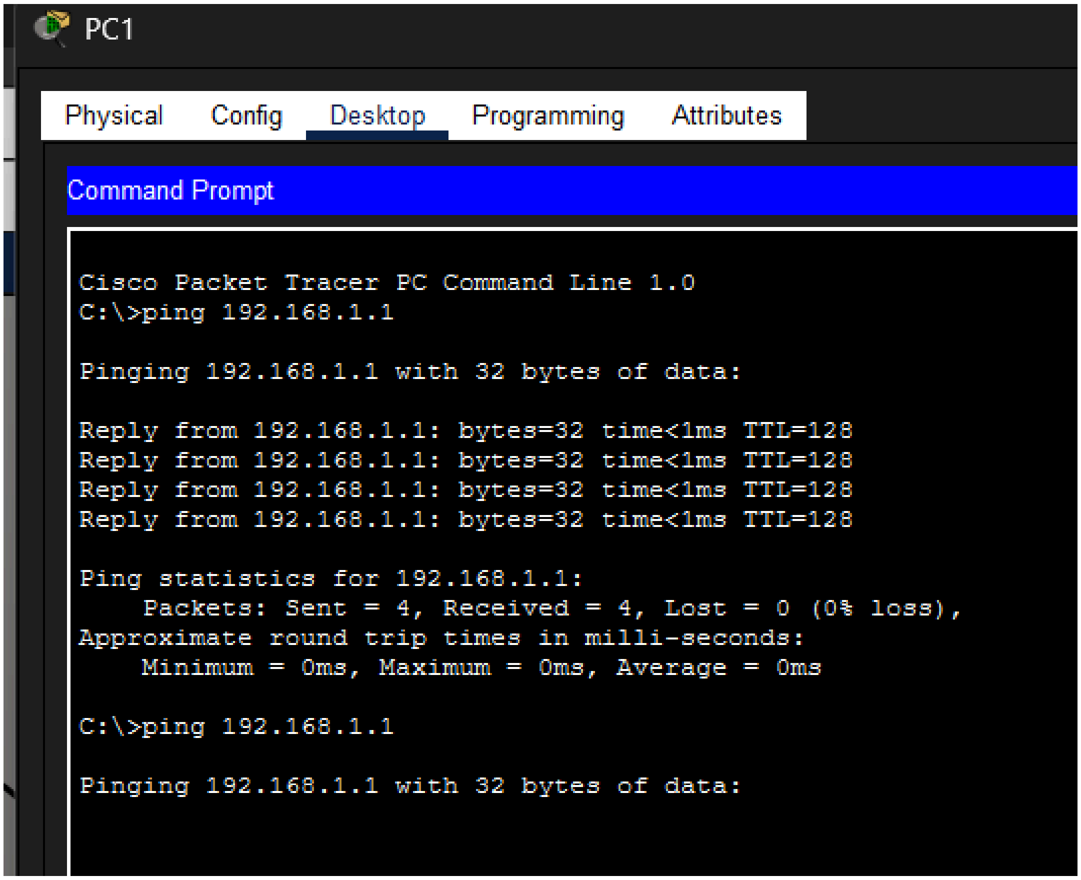


Configuring the vlans to carry data and voice.

Switch0

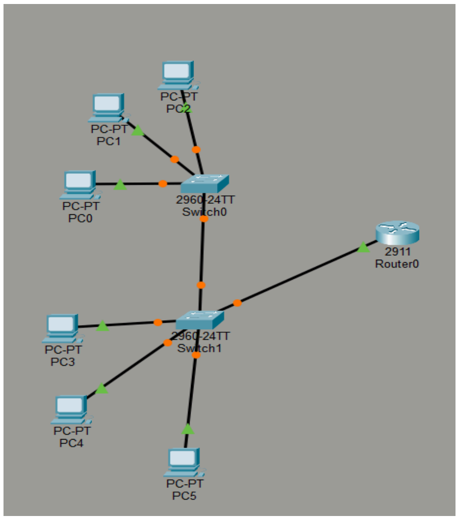
Physical Config CLI Attributes

```
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#do show vlan bri

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   voice                            active    Fa0/1
20   data                             active    Fa0/1
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch(config-if)#do show mac-address table
show mac-address table
                 ^
% Invalid input detected at '^' marker.

Switch(config-if)#do show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

  10    0001.9656.b638    DYNAMIC     Fa0/1
Switch(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch(config-if)#
Switch(config-if)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#do show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

  10    0001.9656.b638    DYNAMIC     Fa0/1
  20    0001.4321.4a6e    DYNAMIC     Fa0/2
  20    0090.2bb4.1ca9    DYNAMIC     Fa0/1
Switch(config-if)#
```




This configuration allows both the IP phone and PC to share the same switch port. The IP phone tags voice traffic for VLAN 10, while the PC sends untagged data traffic that is placed in VLAN 20.

9. Try Inter VLAN routing with Router.
Set up trunk ports between switches and try pinging between different vlans




vlan configuration on switch0

```
Switch#
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

```
%SYS-5-CONFIG_I: Configured from console by console

Switch#show interface status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    10         a-full  a-100 10/100BaseTX
Fa0/2                        connected    10         a-full  a-100 10/100BaseTX
Fa0/3                        connected    10         a-full  a-100 10/100BaseTX
```

```
Switch>enable
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
10   IT                               active    Fa0/1, Fa0/2, Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Setting up the trunk port

```
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface Fa0/24
Switch(config-if)#switchport mode trunk
```

```
Fa0/24                    connected    trunk      a-full  a-100 10/100BaseTX
```

Vlan configuration on switch1

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
20   Fin                              active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface Fa0/24
Switch(config-if)#switchport  mode trunk
Switch(config-if)#exit
Switch(config)#
Switch(config)#exit
```

```
Fa0/24                    connected    trunk      a-full  a-100 10/100BaseTX
```

Enabling the interface in router

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

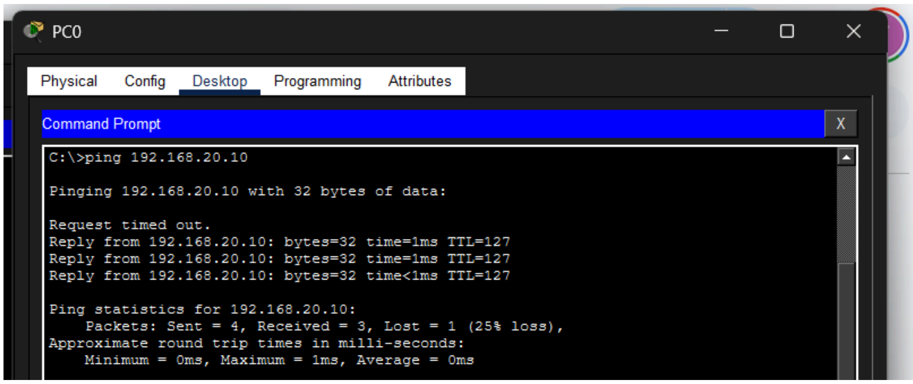
Router on stick architecture configuring the interfaces for vlan

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#interface g0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to
up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#exit
Router(config)#interface g0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to
up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
```

On switch1 shows the trunked ports and vlans allowed on those trunked ports

```
Switch1
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/24      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005
Fa0/24      1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20
Fa0/24      1,10,20

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20
Fa0/24      1,10,20

Switch#
```

Pinging pcs in two different vlans. It has to be routed via the routers.




10. You configured VLANs 10 and 20 on your switch and assigned ports to each VLAN. However, devices in VLAN 10 cannot communicate with devices in VLAN 20. Troubleshoot the issue.

Troubleshooting VLAN Communication Issue (VLAN 10 ↔ VLAN 20)

Devices in different VLANs cannot communicate by default because VLANs create separate broadcast domains. To enable communication between VLAN 10 and VLAN 20, inter-VLAN routing must be configured.

Step 1: Verify VLAN Configuration

show vlan brief

- Ensure VLAN 10 and VLAN 20 exist
- Ensure correct ports are assigned

Step 2: Check Interface Status
show ip interface brief

- Ensure interfaces are up/up

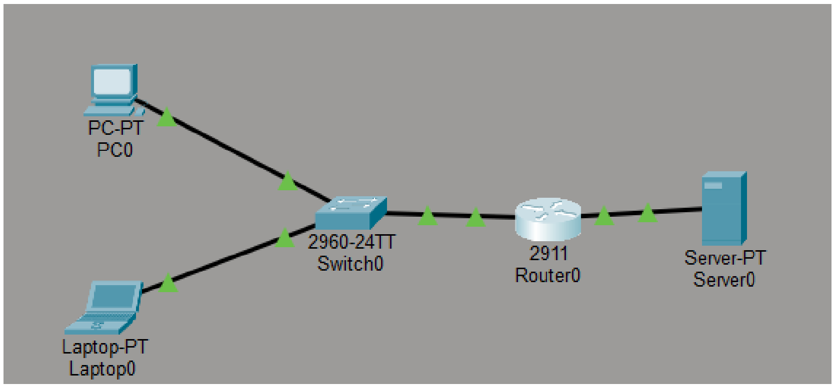
Step 3: Identify Root Cause

- VLANs are isolated at Layer 2
- No routing device is configured
- Therefore, communication between VLANs is not possible

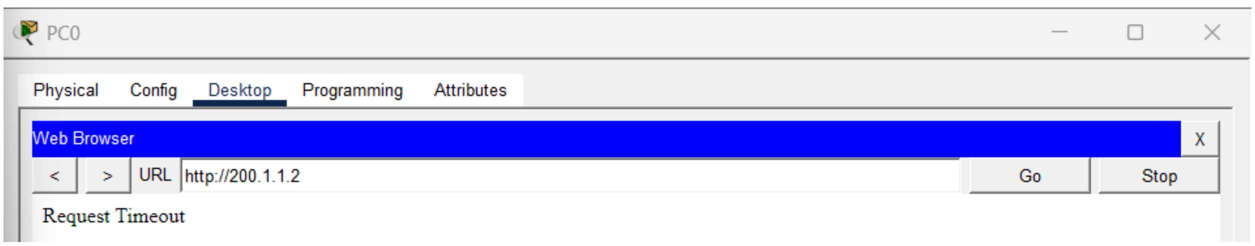
Step 4: Solution – Enable Inter-VLAN Routing

11. Implement ACLs to restrict traffic based on source and destination ports.Test rules by simulating legitimate and unauthorized traffic.

Network topology




FTP and HTTP services fail.



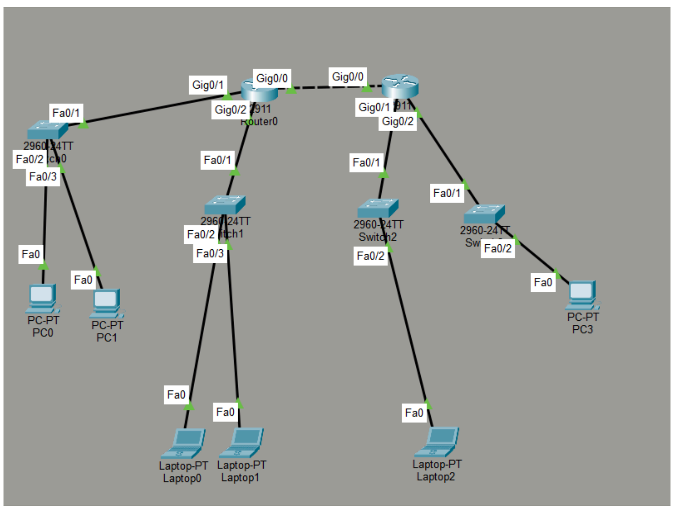


blocking HTTP and FTP services is an example of destination port-based filtering, as it restricts access to services running on specific ports. Source port filtering can also be applied to restrict traffic originating from specific services.

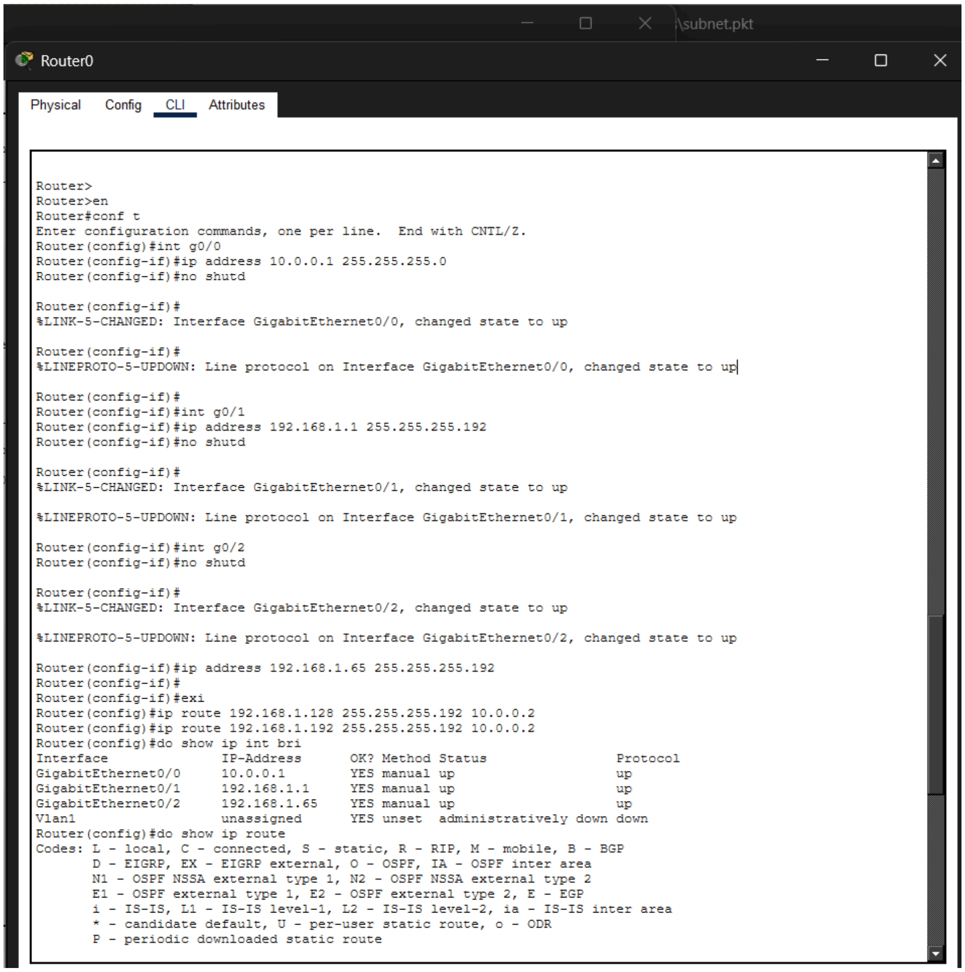
```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#access-list 100 deny tcp any eq 80 any
Router(config)#access-list 100 deny tcp any any eq 80
Router(config)#
```

12. Configure a standard Access Control List (ACL) on a router to permit traffic from a specific IP range.Test connectivity to verify the ACL is working as intended.

Network topology



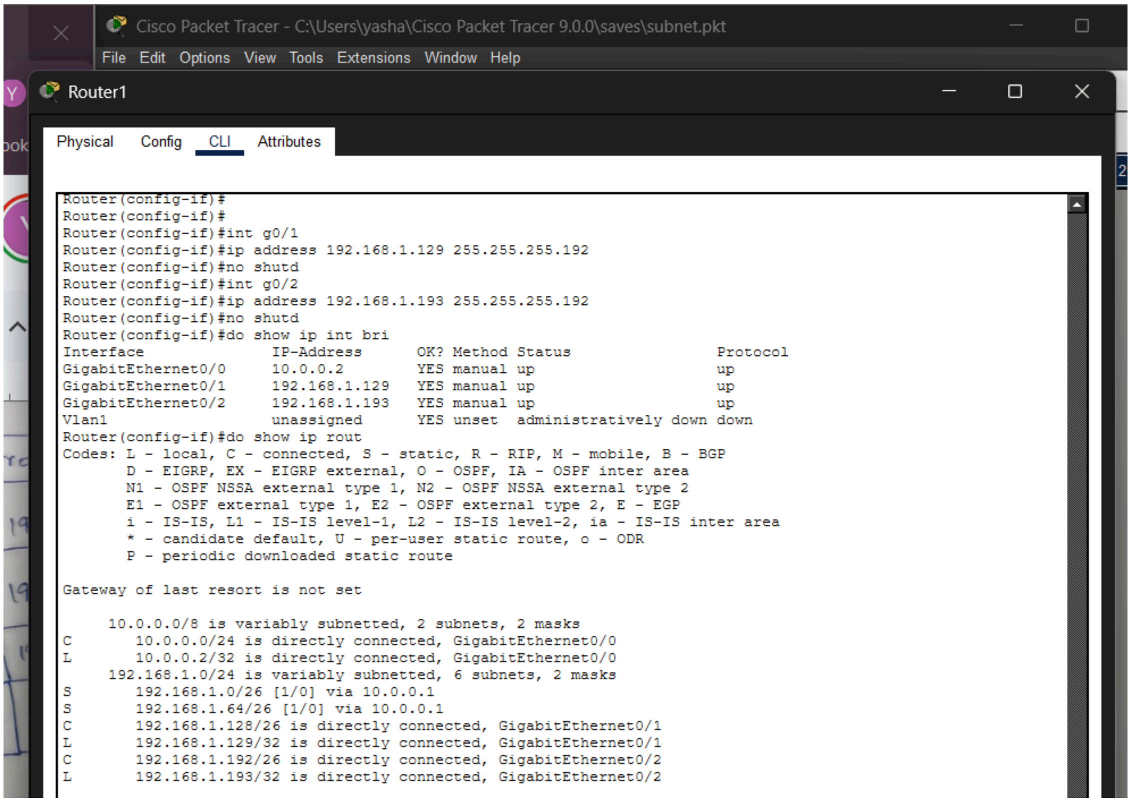
Assigning Router0 interfaces

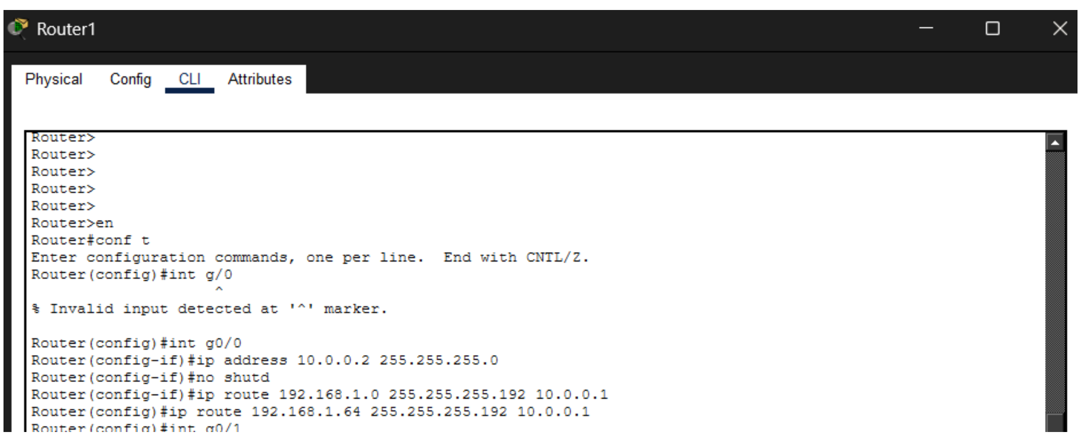



```
Router>
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int g0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.0
Router(config-if)#no shutd

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#int g0/1
Router(config-if)#ip address 192.168.1.1 255.255.255.192
Router(config-if)#no shutd

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#int g0/2
Router(config-if)#no shutd

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Router(config-if)#ip address 192.168.1.65 255.255.255.192
Router(config-if)#
Router(config-if)#exi
Router(config)#ip route 192.168.1.128 255.255.255.192 10.0.0.2
Router(config)#ip route 192.168.1.192 255.255.255.192 10.0.0.2
Router(config)#do show ip int bri
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.0.1        YES manual up                    up
GigabitEthernet0/1     192.168.1.1     YES manual up                    up
GigabitEthernet0/2     192.168.1.65    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

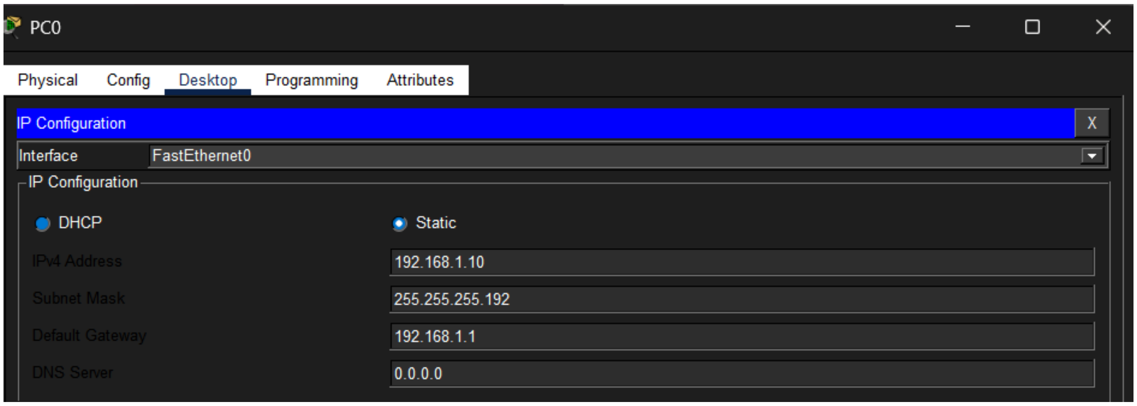
```
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
     192.168.1.0/24 is variably subnetted, 6 subnets, 2 masks
C       192.168.1.0/26 is directly connected, GigabitEthernet0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/1
C       192.168.1.64/26 is directly connected, GigabitEthernet0/2
L       192.168.1.65/32 is directly connected, GigabitEthernet0/2
S       192.168.1.128/26 [1/0] via 10.0.0.2
S       192.168.1.192/26 [1/0] via 10.0.0.2

Router(config)#
```

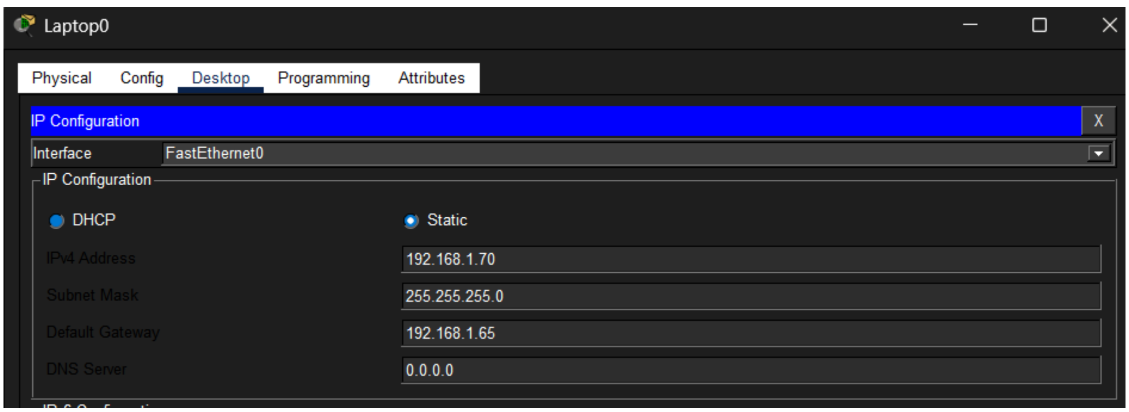
Assigning Router1 interfaces

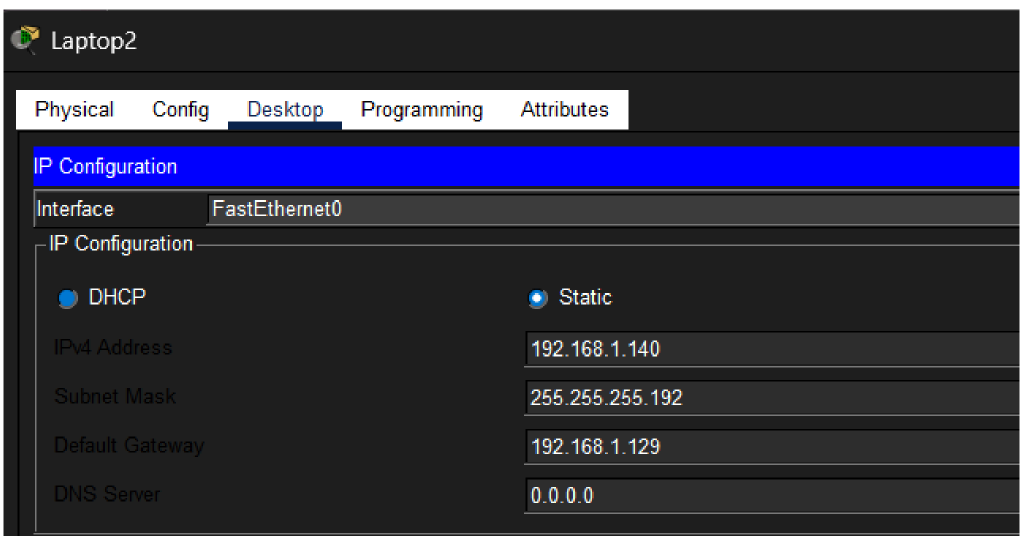


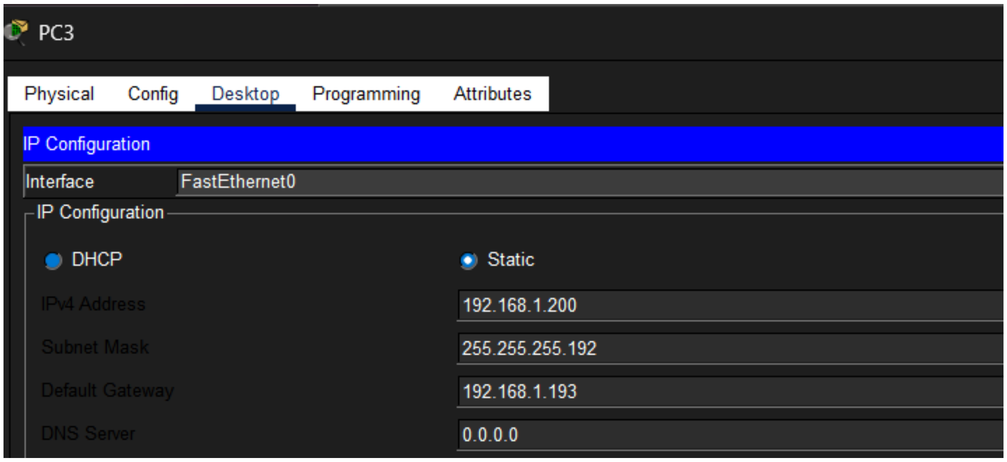



End host configuration

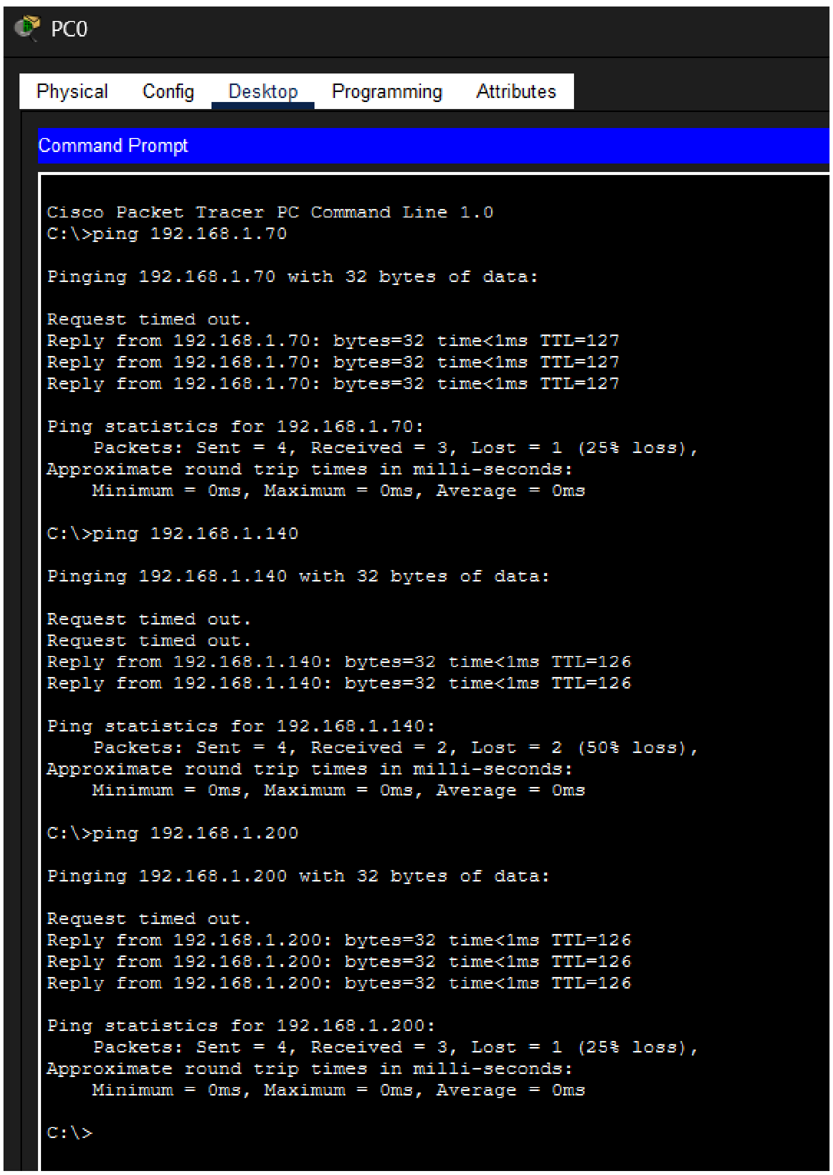


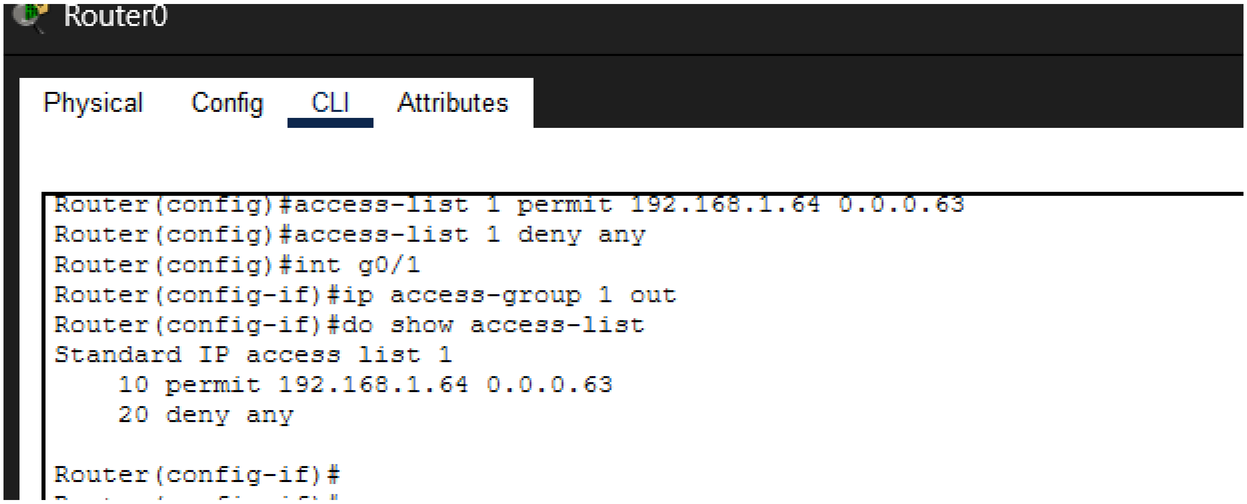



Verifying the connectivity between devices before applying ACL.

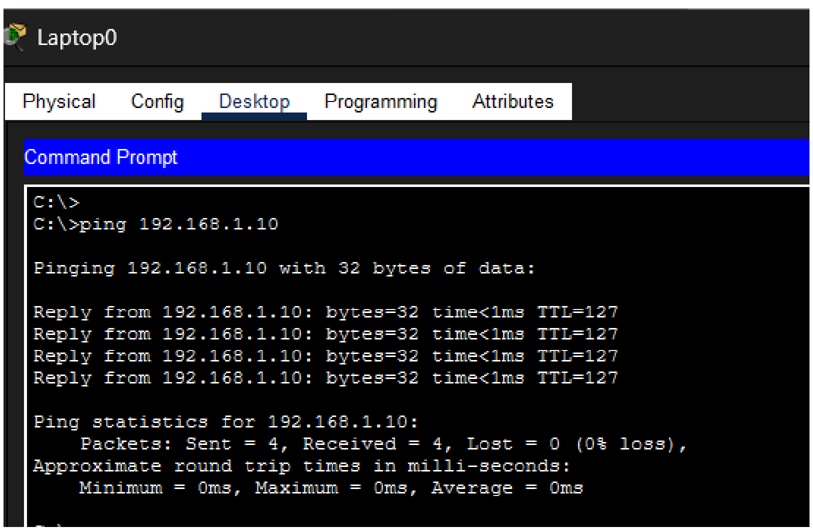


Alllowing the traffic from the 192.168.64.0/26 subnet and blocking the other subnets to communicate with 192.168.1.0/26

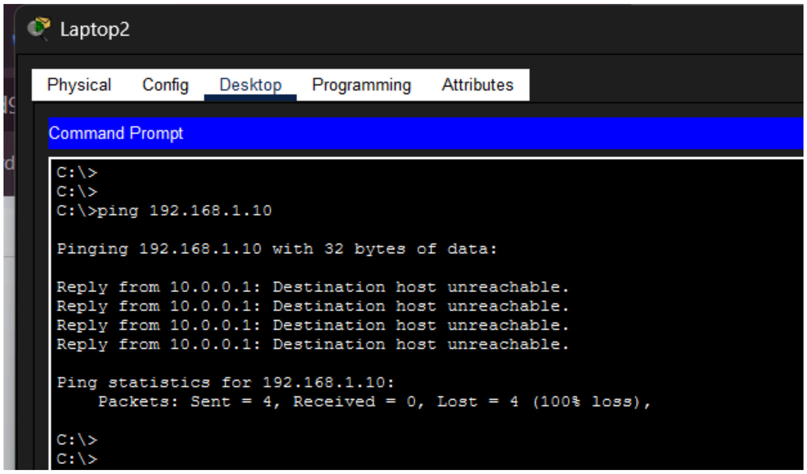
Applying ACL on router0

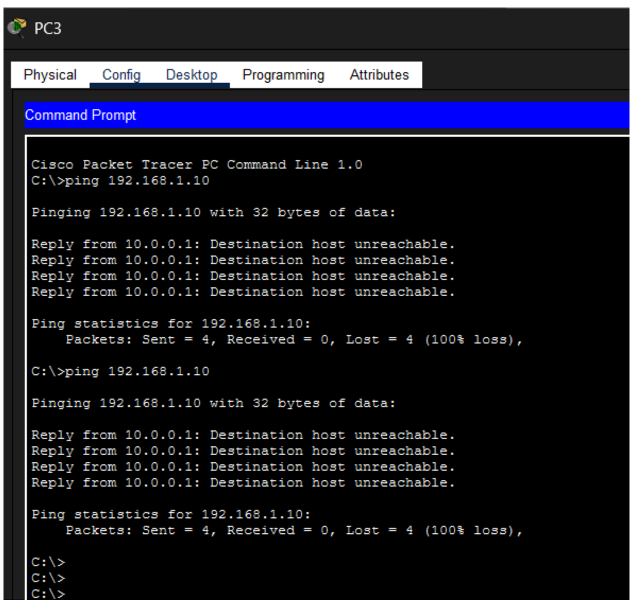



Traffic allowed so we get ping replies



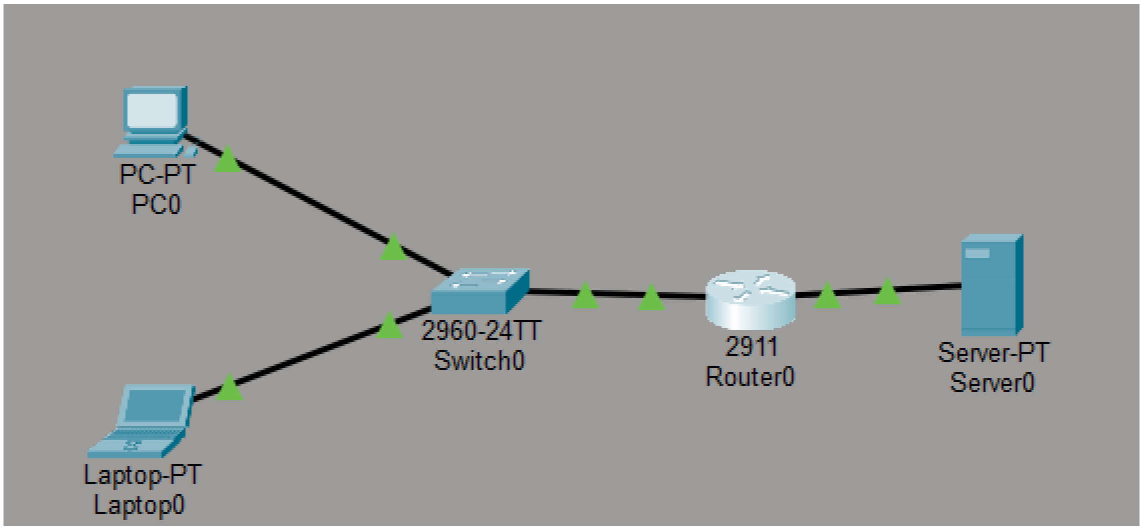
Traffic blocked



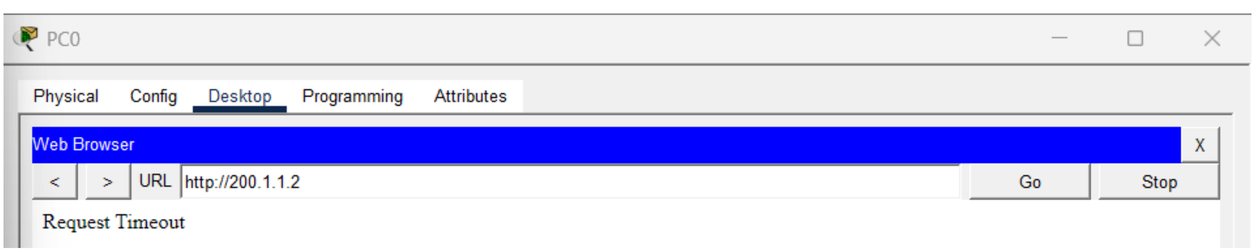
13. Create an extended ACL to block specific applications, such as HTTP or FTP traffic.Test the ACL rules by attempting to access blocked services.

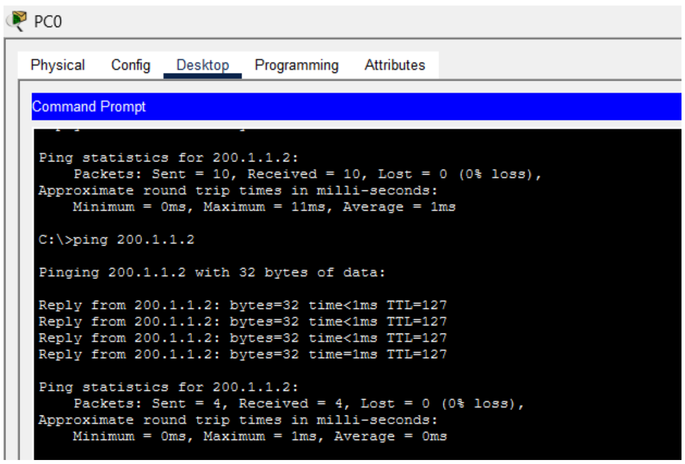
Network topology




```
Router(config)#
Router(config)#
Router(config)#access-list 100 deny tcp any any eq 80
Router(config)#access-list 100 deny tcp any any eq 21
Router(config)#access-list 100 permit ip any any
Router(config)#interface g0/1
Router(config-if)#ip access-group 100 in
Router(config-if)#
```

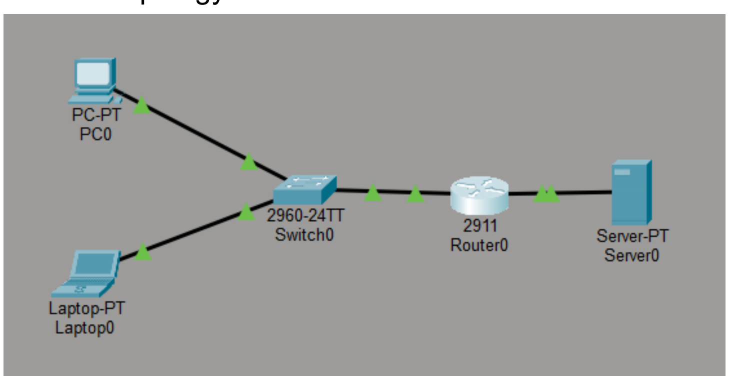
FTP and HTTP services fail but ping works because the ping traffic is allowed by the ACL.



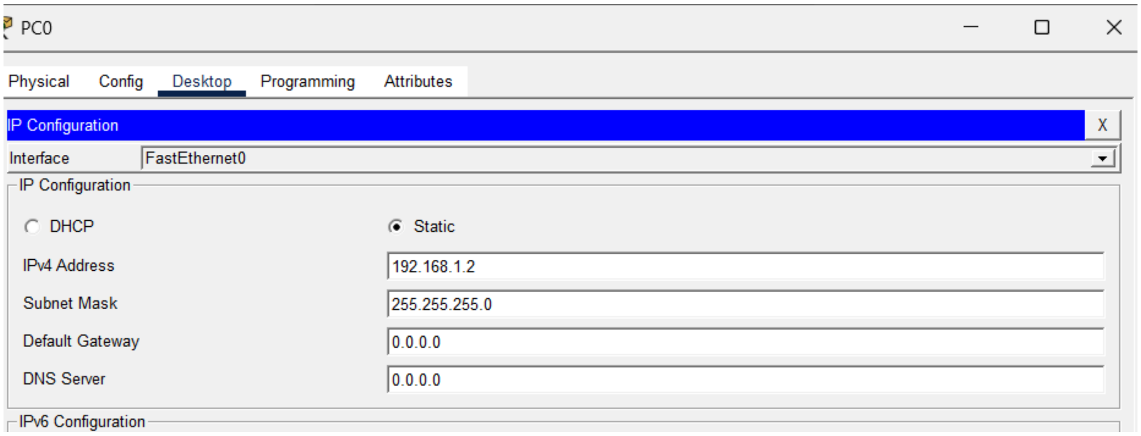



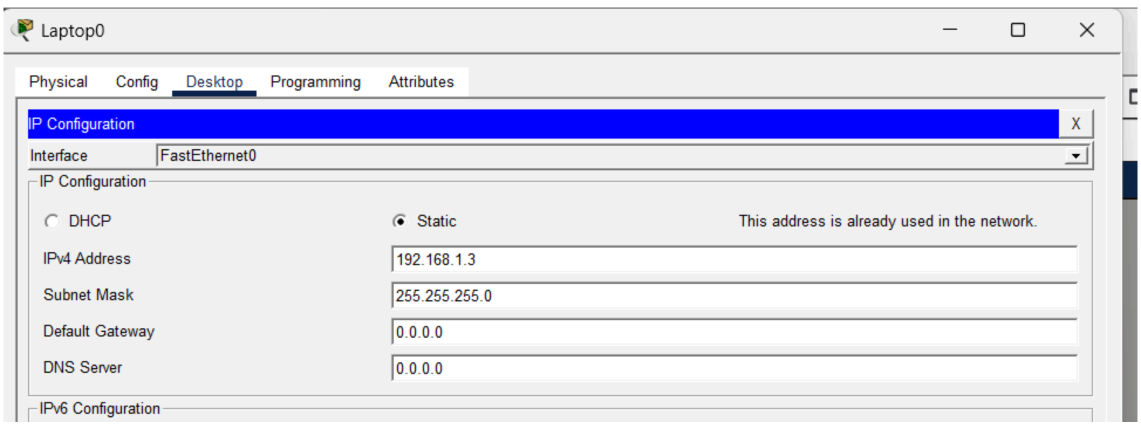

14. Try Static NAT, Dynamic NAT and PAT to translate IPs

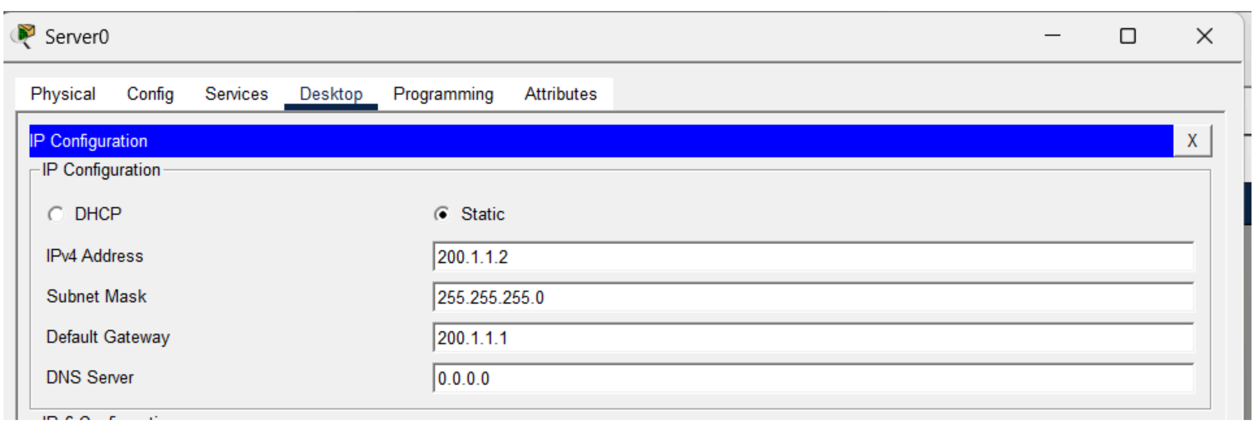
Network topology



IP addresses of the end hosts. The default gateway is changed from 0.0.0.0 to 192.168.1.1.



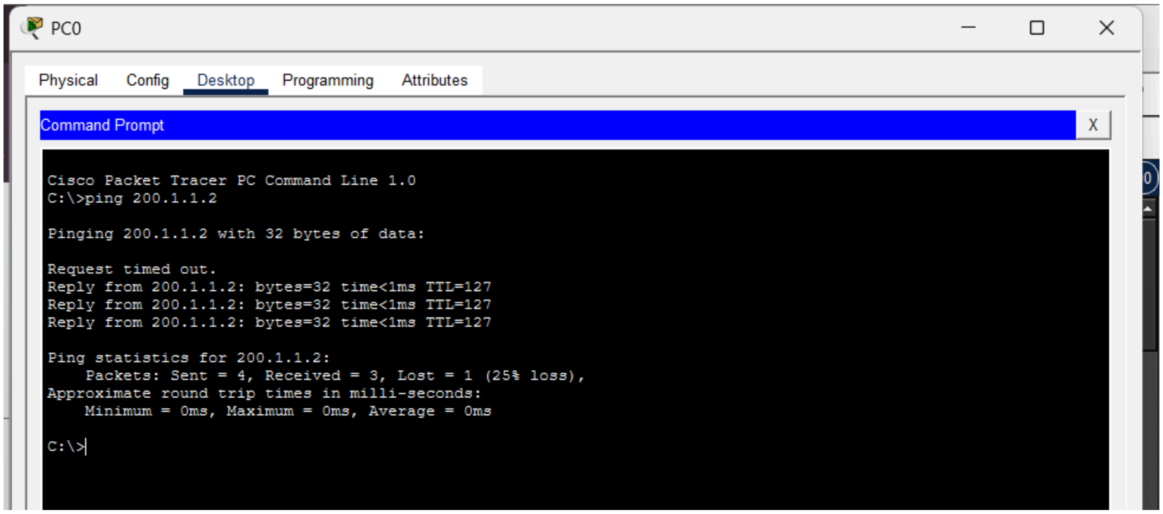
Configuring the interfaces one for the LAN side another for the WAN side

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#interface g0/2
Router(config-if)#ip address 200.1.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Router(config-if)#
```

Shows the interfaces and the routes

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/1
     200.1.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.1.1.0/24 is directly connected, GigabitEthernet0/2
L       200.1.1.1/32 is directly connected, GigabitEthernet0/2
```

Pinging the server before configuring NAT.Before configuring NAT, communication between internal and external network was possible because both networks were directly connected to the router and present in the routing table. NAT is required only when translating private IP addresses for communication with external networks such as the Internet, where private IPs are not routable.




STATIC NAT

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip nat inside source static 192.168.1.2 200.1.1.100
Router(config)#interface g0/1
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#interface g0/2
Router(config-if)#ip nat outside
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#
```

NAT  translations

```
Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 200.1.1.100:21    192.168.1.2:21     200.1.1.2:21       200.1.1.2:21
icmp 200.1.1.100:22    192.168.1.2:22     200.1.1.2:22       200.1.1.2:22
icmp 200.1.1.100:23    192.168.1.2:23     200.1.1.2:23       200.1.1.2:23
icmp 200.1.1.100:24    192.168.1.2:24     200.1.1.2:24       200.1.1.2:24
---  200.1.1.100       192.168.1.2        ---                ---

Router#
```

```
C:\>ping 200.1.1.2

Pinging 200.1.1.2 with 32 bytes of data:

Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time=1ms TTL=127

Ping statistics for 200.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

## DYNAMIC NAT

```
Router(config)#no ip nat inside source static 192.168.1.2 200.1.1.100
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip nat translations
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip nat pool MYPOOL 200.1.1.100 200.1.1.110 netmask 255.255.255.0
Router(config)#access-list permit 192.168.1.0 0.0.0.255
                          ^
% Invalid input detected at '^' marker.

Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#ip nat inside source list 1 pool MYPOOL
Router(config)#exit
Router#
```

Each private ip gets mapped to a public ip.

Router0

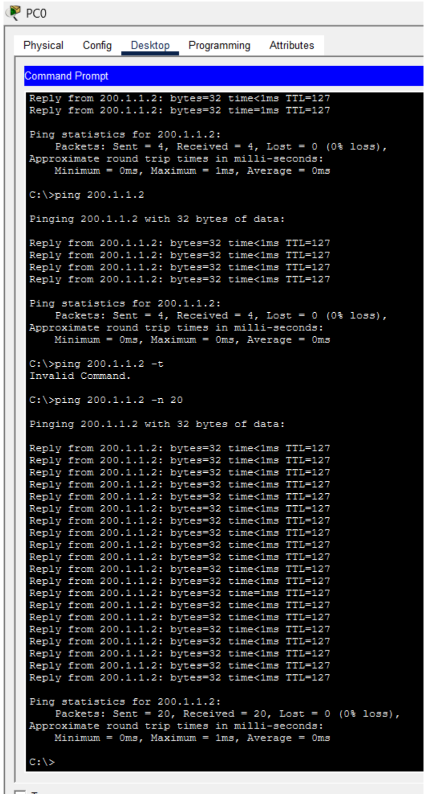
Physical Config CLI Attributes

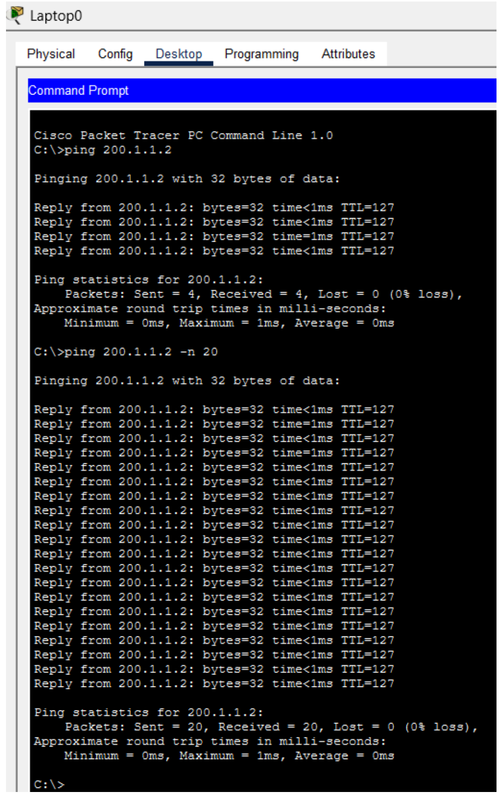
IOS Command Line Interface

```
icmp 200.1.1.100:26    192.168.1.2:26     200.1.1.2:26       200.1.1.2:26
icmp 200.1.1.100:27    192.168.1.2:27     200.1.1.2:27       200.1.1.2:27
icmp 200.1.1.100:28    192.168.1.2:28     200.1.1.2:28       200.1.1.2:28

Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 200.1.1.100:1     192.168.1.3:1      200.1.1.2:1        200.1.1.2:1
icmp 200.1.1.100:2     192.168.1.3:2      200.1.1.2:2        200.1.1.2:2
icmp 200.1.1.100:3     192.168.1.3:3      200.1.1.2:3        200.1.1.2:3
icmp 200.1.1.100:4     192.168.1.3:4      200.1.1.2:4        200.1.1.2:4

Router#clear ip nat translations *
                                ^
% Invalid input detected at '^' marker.

Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 200.1.1.100:29    192.168.1.2:29     200.1.1.2:29       200.1.1.2:29
icmp 200.1.1.100:30    192.168.1.2:30     200.1.1.2:30       200.1.1.2:30
icmp 200.1.1.100:31    192.168.1.2:31     200.1.1.2:31       200.1.1.2:31
icmp 200.1.1.100:32    192.168.1.2:32     200.1.1.2:32       200.1.1.2:32
icmp 200.1.1.100:33    192.168.1.2:33     200.1.1.2:33       200.1.1.2:33
icmp 200.1.1.100:34    192.168.1.2:34     200.1.1.2:34       200.1.1.2:34
icmp 200.1.1.100:35    192.168.1.2:35     200.1.1.2:35       200.1.1.2:35
icmp 200.1.1.100:36    192.168.1.2:36     200.1.1.2:36       200.1.1.2:36
icmp 200.1.1.100:37    192.168.1.2:37     200.1.1.2:37       200.1.1.2:37
icmp 200.1.1.100:38    192.168.1.2:38     200.1.1.2:38       200.1.1.2:38
icmp 200.1.1.100:39    192.168.1.2:39     200.1.1.2:39       200.1.1.2:39
icmp 200.1.1.100:40    192.168.1.2:40     200.1.1.2:40       200.1.1.2:40
icmp 200.1.1.100:41    192.168.1.2:41     200.1.1.2:41       200.1.1.2:41
icmp 200.1.1.100:42    192.168.1.2:42     200.1.1.2:42       200.1.1.2:42
icmp 200.1.1.100:43    192.168.1.2:43     200.1.1.2:43       200.1.1.2:43
icmp 200.1.1.100:44    192.168.1.2:44     200.1.1.2:44       200.1.1.2:44
icmp 200.1.1.100:45    192.168.1.2:45     200.1.1.2:45       200.1.1.2:45
icmp 200.1.1.100:46    192.168.1.2:46     200.1.1.2:46       200.1.1.2:46
icmp 200.1.1.100:47    192.168.1.2:47     200.1.1.2:47       200.1.1.2:47
icmp 200.1.1.100:48    192.168.1.2:48     200.1.1.2:48       200.1.1.2:48
icmp 200.1.1.101:10    192.168.1.3:10     200.1.1.2:10       200.1.1.2:10
icmp 200.1.1.101:11    192.168.1.3:11     200.1.1.2:11       200.1.1.2:11
icmp 200.1.1.101:12    192.168.1.3:12     200.1.1.2:12       200.1.1.2:12
icmp 200.1.1.101:13    192.168.1.3:13     200.1.1.2:13       200.1.1.2:13
icmp 200.1.1.101:14    192.168.1.3:14     200.1.1.2:14       200.1.1.2:14
icmp 200.1.1.101:15    192.168.1.3:15     200.1.1.2:15       200.1.1.2:15
icmp 200.1.1.101:16    192.168.1.3:16     200.1.1.2:16       200.1.1.2:16
icmp 200.1.1.101:17    192.168.1.3:17     200.1.1.2:17       200.1.1.2:17
icmp 200.1.1.101:5     192.168.1.3:5      200.1.1.2:5        200.1.1.2:5
icmp 200.1.1.101:6     192.168.1.3:6      200.1.1.2:6        200.1.1.2:6
icmp 200.1.1.101:7     192.168.1.3:7      200.1.1.2:7        200.1.1.2:7
icmp 200.1.1.101:8     192.168.1.3:8      200.1.1.2:8        200.1.1.2:8
icmp 200.1.1.101:9     192.168.1.3:9      200.1.1.2:9        200.1.1.2:9

Router#
Router#
Router#
```

Copy Paste

Top




PAT

Many private ips share one public ip.

```
Router(config)#ip nat inside source list 1 interface g0/2 overload
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip nat translations
Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 200.1.1.1:45      192.168.1.3:45     200.1.1.2:45       200.1.1.2:45
icmp 200.1.1.1:46      192.168.1.3:46     200.1.1.2:46       200.1.1.2:46
icmp 200.1.1.1:47      192.168.1.3:47     200.1.1.2:47       200.1.1.2:47
icmp 200.1.1.1:48      192.168.1.3:48     200.1.1.2:48       200.1.1.2:48
icmp 200.1.1.1:49      192.168.1.3:49     200.1.1.2:49       200.1.1.2:49
icmp 200.1.1.1:50      192.168.1.3:50     200.1.1.2:50       200.1.1.2:50
icmp 200.1.1.1:51      192.168.1.3:51     200.1.1.2:51       200.1.1.2:51
icmp 200.1.1.1:52      192.168.1.3:52     200.1.1.2:52       200.1.1.2:52
icmp 200.1.1.1:53      192.168.1.3:53     200.1.1.2:53       200.1.1.2:53
icmp 200.1.1.1:54      192.168.1.3:54     200.1.1.2:54       200.1.1.2:54
icmp 200.1.1.1:69      192.168.1.2:69     200.1.1.2:69       200.1.1.2:69
icmp 200.1.1.1:70      192.168.1.2:70     200.1.1.2:70       200.1.1.2:70
icmp 200.1.1.1:71      192.168.1.2:71     200.1.1.2:71       200.1.1.2:71
icmp 200.1.1.1:72      192.168.1.2:72     200.1.1.2:72       200.1.1.2:72
icmp 200.1.1.1:73      192.168.1.2:73     200.1.1.2:73       200.1.1.2:73
icmp 200.1.1.1:74      192.168.1.2:74     200.1.1.2:74       200.1.1.2:74
icmp 200.1.1.1:75      192.168.1.2:75     200.1.1.2:75       200.1.1.2:75
icmp 200.1.1.1:76      192.168.1.2:76     200.1.1.2:76       200.1.1.2:76
icmp 200.1.1.1:77      192.168.1.2:77     200.1.1.2:77       200.1.1.2:77
icmp 200.1.1.1:78      192.168.1.2:78     200.1.1.2:78       200.1.1.2:78

Router#
```

15.Download iperf in laptop/phone and make sure they are in same network. Try different iperf commands with tcp, udp, birectional, reverse, multicast, parallel options and analyze the bandwidth and rate of transmission, delay, jitter etc.

Ipperf3 has to be downloaded as a zip file and extracted.

```
PS E:\> cd Downloads
PS E:\Downloads> cd "iperf3.1.4_64"
PS E:\Downloads\iperf3.1.4_64> ls


    Directory: E:\Downloads\iperf3.1.4_64


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        22-03-2026     10:00        3306962 cygwin1.dll
-a----        22-03-2026     10:00         472387 iperf3.exe
```

```
Administrator: Windows PowerShell
PS E:\Downloads\iperf3.1.4_64> iperf3.exe
iperf3.exe : The term 'iperf3.exe' is not recognized as the name of a cmdlet, function, script file,
or operable program. Check the spelling of the name, or if a path was included, verify that the path
is correct and try again.
At line:1 char:1
+ iperf3.exe
+ ~~~~~~~~~~
    + CategoryInfo          : ObjectNotFound: (iperf3.exe:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException


Suggestion [3,General]: The command iperf3.exe was not found, but does exist in the current location. Wi
ndows PowerShell does not load commands from the current location by default. If you trust this command,
 instead type: ".\iperf3.exe". See "get-help about_Command_Precedence" for more details.
PS E:\Downloads\iperf3.1.4_64> .\iperf3.exe
iperf3: parameter error - must either be a client (-c) or server (-s)

Usage: iperf [-s|-c host] [options]
       iperf [-h|--help] [-v|--version]

Server or Client:
  -p, --port      #         server port to listen on/connect to
  -f, --format    [kmgKMG]  format to report: Kbits, Mbits, KBytes, MBytes
  -i, --interval  #         seconds between periodic bandwidth reports
  -F, --file name           xmit/recv the specified file
  -B, --bind      <host>    bind to a specific interface
  -V, --verbose             more detailed output
  -J, --json                output in JSON format
  --logfile f               send output to a log file
  -d, --debug               emit debugging output
  -v, --version             show version information and quit
  -h, --help                show this message and quit
Server specific:
  -s, --server              run in server mode
  -D, --daemon              run the server as a daemon
  -I, --pidfile file        write PID file
  -1, --one-off             handle one client connection then exit
Client specific:
  -c, --client    <host>    run in client mode, connecting to <host>
  -u, --udp                 use UDP rather than TCP
  -b, --bandwidth #[KMG][/#] target bandwidth in bits/sec (0 for unlimited)
                            (default 1 Mbit/sec for UDP, unlimited for TCP)
                            (optional slash and packet count for burst mode)
  -t, --time      #         time in seconds to transmit for (default 10 secs)
  -n, --bytes     #[KMG]    number of bytes to transmit (instead of -t)
  -k, --blockcount #[KMG]   number of blocks (packets) to transmit (instead of -t or -n)
  -l, --len       #[KMG]    length of buffer to read or write
                            (default 128 KB for TCP, 8 KB for UDP)
  --cport         <port>    bind to a specific client port (TCP and UDP, default: ephemeral port)
```

The laptop and phone need to be in the same network.The laptop acts as the server and the phone acts as the client.

The server is started on the laptop and the iperf3 on the phone must be configured wth the corresponding ip address to see the connectivity metrics.

```
PS E:\Downloads\iperf3.1.4_64> .\iperf3 -s
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
Accepted connection from 172.28.128.67, port 53534
[  5] local 172.28.128.153 port 5201 connected to 172.28.128.67 port 53540
[ ID] Interval           Transfer     Bandwidth
[  5]   0.00-1.01   sec  3.62 MBytes  30.1 Mbits/sec
[  5]   1.01-2.01   sec  2.12 MBytes  17.8 Mbits/sec
[  5]   2.01-3.00   sec  5.88 MBytes  49.7 Mbits/sec
[  5]   3.00-4.00   sec  6.88 MBytes  57.5 Mbits/sec
[  5]   4.00-5.00   sec  1.75 MBytes  14.7 Mbits/sec
[  5]   5.00-6.02   sec  3.62 MBytes  30.0 Mbits/sec
[  5]   6.02-7.00   sec  4.62 MBytes  39.4 Mbits/sec
[  5]   7.00-8.00   sec  7.62 MBytes  63.9 Mbits/sec
[  5]   8.00-9.01   sec  5.38 MBytes  44.6 Mbits/sec
[  5]   9.01-10.00  sec  11.0 MBytes  93.2 Mbits/sec
[  5]  10.00-10.01  sec   128 KBytes   114 Mbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth
[  5]   0.00-10.01  sec  52.6 MBytes  44.1 Mbits/sec                  sender
[  5]   0.00-10.01  sec  0.00 Bytes  0.00 bits/sec                  receiver
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
```

Using the the same system (laptop) to test out the metrics while the UDP protocol is used.

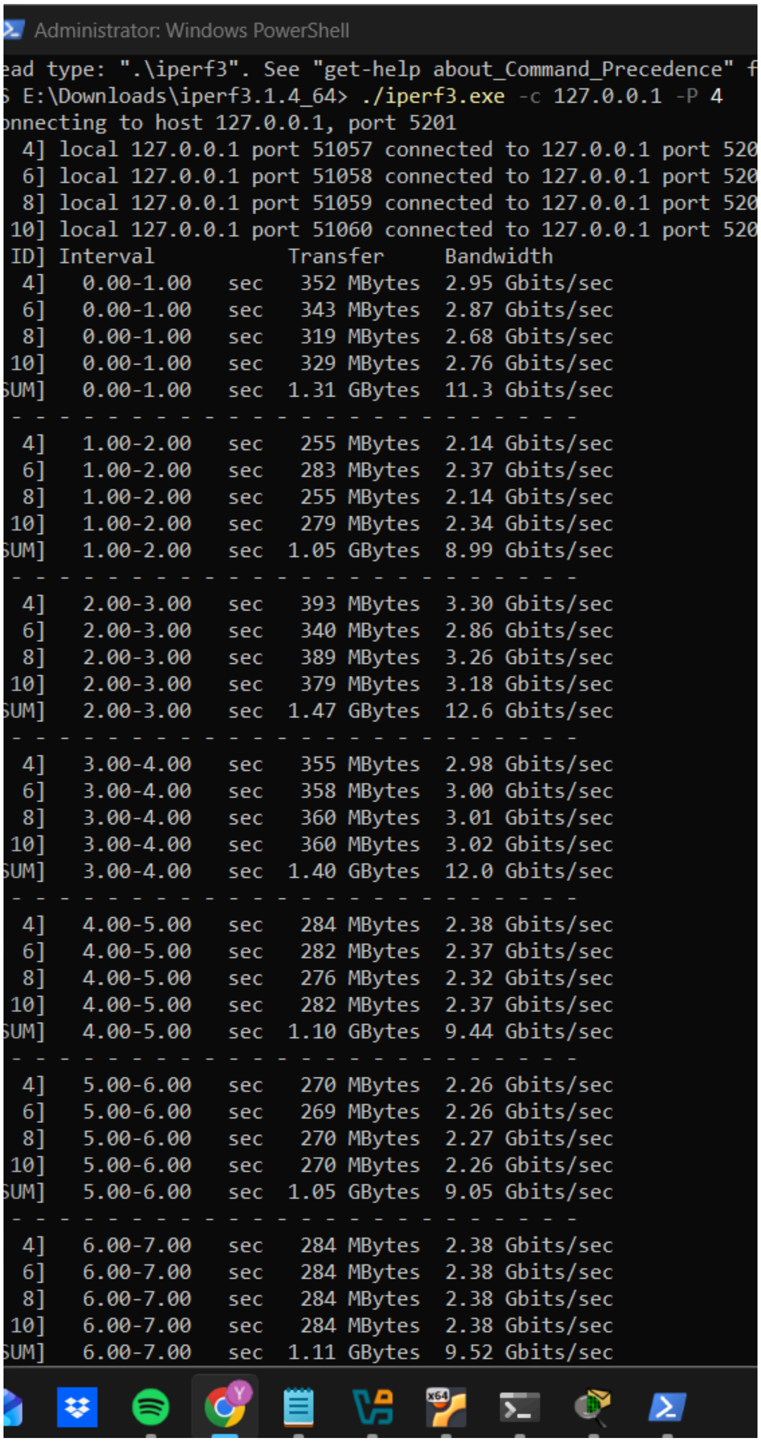
```
PS E:\Downloads\iperf3.1.4_64> ./iperf3.exe -c 127.0.0.1 -u -b 100M
Connecting to host 127.0.0.1, port 5201
[  4] local 127.0.0.1 port 56643 connected to 127.0.0.1 port 5201
[ ID] Interval           Transfer     Bandwidth       Total Datagrams
[  4]   0.00-1.00   sec  10.9 MBytes  91.4 Mbits/sec  1396
[  4]   1.00-2.00   sec  11.8 MBytes  99.3 Mbits/sec  1515
[  4]   2.00-3.00   sec  12.0 MBytes   100 Mbits/sec  1531
[  4]   3.00-4.00   sec  11.9 MBytes   100 Mbits/sec  1525
[  4]   4.00-5.00   sec  11.9 MBytes  99.8 Mbits/sec  1523
[  4]   5.00-6.00   sec  11.9 MBytes   100 Mbits/sec  1527
[  4]   6.00-7.00   sec  11.9 MBytes   100 Mbits/sec  1526
[  4]   7.00-8.00   sec  11.9 MBytes  99.9 Mbits/sec  1524
[  4]   8.00-9.00   sec  11.9 MBytes   100 Mbits/sec  1526
[  4]   9.00-10.00  sec  12.0 MBytes   100 Mbits/sec  1530
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth       Jitter    Lost/Total Datagrams
[  4]   0.00-10.00  sec   118 MBytes  99.1 Mbits/sec  0.019 ms  20/15123 (0.13%)
[  4] Sent 15123 datagrams

iperf Done.
PS E:\Downloads\iperf3.1.4_64>
```

Reverse Mode

```
PS E:\Downloads\iperf3.1.4_64> ./iperf3.exe -c 127.0.0.1 -R
Connecting to host 127.0.0.1, port 5201
Reverse mode, remote host 127.0.0.1 is sending
[  4] local 127.0.0.1 port 55699 connected to 127.0.0.1 port 5201
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-1.00   sec  1.65 GBytes  14.1 Gbits/sec
[  4]   1.00-2.00   sec  1.17 GBytes  10.0 Gbits/sec
[  4]   2.00-3.00   sec  1.35 GBytes  11.6 Gbits/sec
[  4]   3.00-4.00   sec  1.45 GBytes  12.4 Gbits/sec
[  4]   4.00-5.00   sec   909 MBytes  7.62 Gbits/sec
[  4]   5.00-6.00   sec  1.33 GBytes  11.4 Gbits/sec
[  4]   6.00-7.00   sec  1.06 GBytes  9.08 Gbits/sec
[  4]   7.00-8.00   sec  1.45 GBytes  12.5 Gbits/sec
[  4]   8.00-9.00   sec  1.19 GBytes  10.2 Gbits/sec
[  4]   9.00-10.00  sec   978 MBytes  8.20 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-10.00  sec  12.5 GBytes  10.7 Gbits/sec                  sender
[  4]   0.00-10.00  sec  12.5 GBytes  10.7 Gbits/sec                  receiver

iperf Done.
```

Parallel streams

```
- - - - - - - - - - - - - - - - - - - - - - - - -
[  4]   7.00-8.00   sec   212 MBytes  1.78 Gbits/sec
[  6]   7.00-8.00   sec   209 MBytes  1.75 Gbits/sec
[  8]   7.00-8.00   sec   145 MBytes  1.22 Gbits/sec
[ 10]   7.00-8.00   sec   212 MBytes  1.78 Gbits/sec
[SUM]   7.00-8.00   sec   778 MBytes  6.52 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[  4]   8.00-9.00   sec   210 MBytes  1.76 Gbits/sec
[  6]   8.00-9.00   sec   223 MBytes  1.87 Gbits/sec
[  8]   8.00-9.00   sec   223 MBytes  1.87 Gbits/sec
[ 10]   8.00-9.00   sec   208 MBytes  1.74 Gbits/sec
[SUM]   8.00-9.00   sec   864 MBytes  7.24 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[  4]   9.00-10.00  sec   167 MBytes  1.40 Gbits/sec
[  6]   9.00-10.00  sec   170 MBytes  1.43 Gbits/sec
[  8]   9.00-10.00  sec   170 MBytes  1.43 Gbits/sec
[ 10]   9.00-10.00  sec   100 MBytes   843 Mbits/sec
[SUM]   9.00-10.00  sec   607 MBytes  5.10 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-10.00  sec  2.72 GBytes  2.33 Gbits/sec                  sender
[  4]   0.00-10.00  sec  2.72 GBytes  2.33 Gbits/sec                  receive
[  6]   0.00-10.00  sec  2.70 GBytes  2.32 Gbits/sec                  sender
[  6]   0.00-10.00  sec  2.70 GBytes  2.32 Gbits/sec                  receive
[  8]   0.00-10.00  sec  2.63 GBytes  2.26 Gbits/sec                  sender
[  8]   0.00-10.00  sec  2.63 GBytes  2.26 Gbits/sec                  receive
[ 10]   0.00-10.00  sec  2.64 GBytes  2.27 Gbits/sec                  sender
[ 10]   0.00-10.00  sec  2.64 GBytes  2.27 Gbits/sec                  receive
[SUM]   0.00-10.00  sec  10.7 GBytes  9.17 Gbits/sec                  sender
[SUM]   0.00-10.00  sec  10.7 GBytes  9.17 Gbits/sec                  receive

iperf Done.
```


Administrator: Windows PowerShell
ead type: ".\iperf3". See "get-help about_Command_Precedence" f
S E:\Downloads\iperf3.1.4_64> ./iperf3.exe -c 127.0.0.1 -P 4
onnecting to host 127.0.0.1, port 5201
4] local 127.0.0.1 port 51057 connected to 127.0.0.1 port 520
6] local 127.0.0.1 port 51058 connected to 127.0.0.1 port 520
8] local 127.0.0.1 port 51059 connected to 127.0.0.1 port 520
10] local 127.0.0.1 port 51060 connected to 127.0.0.1 port 520
ID] Interval Transfer Bandwidth
4] 0.00-1.00 sec 352 MBytes 2.95 Gbits/sec
6] 0.00-1.00 sec 343 MBytes 2.87 Gbits/sec
8] 0.00-1.00 sec 319 MBytes 2.68 Gbits/sec
10] 0.00-1.00 sec 329 MBytes 2.76 Gbits/sec
SUM] 0.00-1.00 sec 1.31 GBytes 11.3 Gbits/sec
4] 1.00-2.00 sec 255 MBytes 2.14 Gbits/sec
6] 1.00-2.00 sec 283 MBytes 2.37 Gbits/sec
8] 1.00-2.00 sec 255 MBytes 2.14 Gbits/sec
10] 1.00-2.00 sec 279 MBytes 2.34 Gbits/sec
SUM] 1.00-2.00 sec 1.05 GBytes 8.99 Gbits/sec
4] 2.00-3.00 sec 393 MBytes 3.30 Gbits/sec
6] 2.00-3.00 sec 340 MBytes 2.86 Gbits/sec
8] 2.00-3.00 sec 389 MBytes 3.26 Gbits/sec
10] 2.00-3.00 sec 379 MBytes 3.18 Gbits/sec
SUM] 2.00-3.00 sec 1.47 GBytes 12.6 Gbits/sec
4] 3.00-4.00 sec 355 MBytes 2.98 Gbits/sec
6] 3.00-4.00 sec 358 MBytes 3.00 Gbits/sec
8] 3.00-4.00 sec 360 MBytes 3.01 Gbits/sec
10] 3.00-4.00 sec 360 MBytes 3.02 Gbits/sec
SUM] 3.00-4.00 sec 1.40 GBytes 12.0 Gbits/sec
4] 4.00-5.00 sec 284 MBytes 2.38 Gbits/sec
6] 4.00-5.00 sec 282 MBytes 2.37 Gbits/sec
8] 4.00-5.00 sec 276 MBytes 2.32 Gbits/sec
10] 4.00-5.00 sec 282 MBytes 2.37 Gbits/sec
SUM] 4.00-5.00 sec 1.10 GBytes 9.44 Gbits/sec
4] 5.00-6.00 sec 270 MBytes 2.26 Gbits/sec
6] 5.00-6.00 sec 269 MBytes 2.26 Gbits/sec
8] 5.00-6.00 sec 270 MBytes 2.27 Gbits/sec
10] 5.00-6.00 sec 270 MBytes 2.26 Gbits/sec
SUM] 5.00-6.00 sec 1.05 GBytes 9.05 Gbits/sec
4] 6.00-7.00 sec 284 MBytes 2.38 Gbits/sec
6] 6.00-7.00 sec 284 MBytes 2.38 Gbits/sec
8] 6.00-7.00 sec 284 MBytes 2.38 Gbits/sec
10] 6.00-7.00 sec 284 MBytes 2.38 Gbits/sec
SUM] 6.00-7.00 sec 1.11 GBytes 9.52 Gbits/sec